

ISC Class XII Mathematics

Chapter 8

Applications of Integrals

28 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Types Covered

- I. Area under $y^2=kx$ / $x^2=ky$ forms between two lines
- II. Area under $y=f(x)$, a polynomial curve
- III. Area under a straight line (trapezium via integration)
- IV. Area under $y=k\sqrt{x}$
- V. Area bounded by $x=c-y^2$ and the y-axis
- VI. Area under a parabola cut by its latus rectum
- VII. Area between a parabola and a chord through the origin
- VIII. Area between $y=x$ and $y=x^2$
- IX. Area of a polygonal region via integration
- X. Area between a parabola and a line – mixed forms
- XI. Area between two parabolas
- XII. Area under/enclosed by a semicircle
- XIII. Area under/enclosed by an ellipse
- XIV. Area of the minor segment of a circle
- XV. Sector-type region: circle + line + axis
- XVI. Smaller region common to an ellipse and a line
- XVII. Segment of an ellipse (arc + chord)
- XVIII. Region common to a circle and another curve
- XIX. Region common to two circles
- XX. Regions bounded by modulus-function curves
- XXI. Area bounded by $y=x|x|$
- XXII. Area under exponential / logarithmic curves
- XXIII. Region defined directly by set-builder inequalities
- XXIV. Equal-area division – find the unknown
- XXV. Ratio of two areas, same curve family
- XXVI. Region bounded by $y=x^2+2$ and two lines
- XXVII. Region bounded by $y=-x^2$ and a line
- XXVIII. ★ Composite region: circle + $y=\sin x$ (4-part)

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Type	Method / Pattern	★	Source(s)
I	Area under $y^2=kx$ / $x^2=ky$ between two lines	Yes	Ex 8, Q1,2,3(i)
II	Area under $y=f(x)$, polynomial	–	Ex 8, Q3(ii)
III	Area under a straight line	–	Ex 8, Q4, Q34
IV	Area under $y=k\sqrt{x}$	Yes	Ex 8, Q5, Q6 (alt.)
V	Area bounded by $x=c-y^2$, y-axis	–	Ex 8, Q7
VI	Parabola cut by latus rectum	–	Ex 8, Q8
VII	Parabola & chord $y=mx$	–	Ex 8, Q9
VIII	Between $y=x$ and $y=x^2$	–	Ex 8, Q10
IX	Polygonal region via integration	Yes	Ex 8, Q11,12,13
X	Parabola & line – mixed forms	–	Ex 8, Q23,24
XI	Between two parabolas	–	Ex 8, Q25
XII	Semicircle area	–	Ex 8, Q14, Q15
XIII	Ellipse area	–	Ex 8, Q16, Q17
XIV	Minor segment of a circle	–	Ex 8, Q18
XV	Sector region: circle+line+axis	–	Ex 8, Q19, Q20
XVI	Smaller region: ellipse & line	–	Ex 8, Q21
XVII	Segment of an ellipse	–	Ex 8, Q22
XVIII	Circle & another curve	–	Ex 8, Q27, Q28
XIX	Region common to two circles	Yes	Ex 8, Q26
XX	Modulus-function regions	–	Ex 8, Q31, Q32
XXI	$y=x x $	–	Ex 8, Q37
XXII	Exponential / log curves	–	Ex 8, Q35
XXIII	Direct set-builder inequality regions	–	Ex 8, Q30, Q33
XXIV	Equal-area division	–	Ex 8, Q39

Type	Method / Pattern	★	Source(s)
XXV	Ratio of two areas	–	Ex 8, Q36
XXVI	$y=x^2+2$ with two lines	–	Ex 8, Q29
XXVII	$y=-x^2$ with a line	–	Ex 8, Q38
XXVIII★	Circle + $y=\sin x$ composite	Yes	Ex 8, Q40

Formula Sheet

Core Area Formulas

Area under a curve (w.r.t. x)

$$A = \int_a^b y \, dx$$

y in terms of x, curve above the x-axis, x from a to b

Area under a curve (w.r.t. y)

$$A = \int_c^d x \, dy$$

x in terms of y, curve to the right of the y-axis, y from c to d

Area between two curves

$$A = \int_a^b [y_{\text{upper}} - y_{\text{lower}}] \, dx$$

Upper curve minus lower curve, over their common interval

Area when curve dips below axis

$$A = \left| \int_a^c y \, dx \right| + \left| \int_c^b y \, dx \right|$$

c is where the curve crosses the axis; each part taken as positive

Standard Curve Areas

Parabola & latus rectum

$$A = \frac{8}{3}a^2$$

Region cut off by $y^2=4ax$ and the line $x=a$

Circle (full)

$$A = \pi r^2$$

Radius r, centered at origin

Circle (quadrant)

$$A = \frac{\pi r^2}{4}$$

One quarter of a circle of radius r

Ellipse (full)

$$A = \pi ab$$

Semi-axes a (x -direction), b (y -direction)

Ellipse (quadrant)

$$A = \frac{\pi ab}{4}$$

One quarter of the standard ellipse

Standard Integrals Used Repeatedly**Power rule**

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$n \neq -1$

Square-root of $(a^2 - x^2)$ form

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$$

Comes from a circle/ellipse; gives sector + triangle split

Exponential

$$\int e^{kx} dx = \frac{e^{kx}}{k} + C$$

k is a nonzero constant

Log integral (by parts)

$$\int \ln x dx = x \ln x - x + C$$

Take $u = \ln x$, $dv = dx$

Reciprocal $x \neq 0$

$$\int \frac{1}{x} dx = \ln |x| + C$$

Geometry Used for Setting Up Bounds**Equation of a line – two points**

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

Used to find the three side-equations of a triangle from its vertices

Points of intersection

Solve curve equations simultaneously

Gives the x- or y-limits of integration – always find these first

Symmetry shortcut

$$A_{\text{full}} = 2 \times A_{\text{half}} \text{ (or } 4 \times A_{\text{quadrant}})$$

Valid only when the curve/region is genuinely symmetric about the axis used

I. Area Under a Single Curve

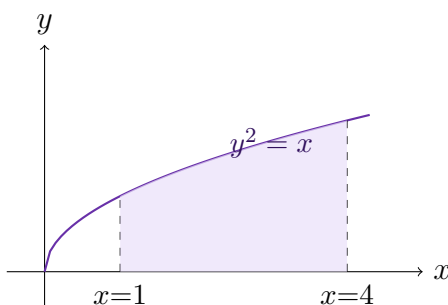
Type 1: Area under $y^2=kx$ / $x^2=ky$ between two lines

The curve is a parabola opening sideways ($y^2=kx$) or upward ($x^2=ky$), and the region is a first-quadrant strip cut off between two parallel lines perpendicular to the axis the parabola opens along. Solve the curve's equation for the "outer" variable, then integrate that expression between the two given bounds. This is the single most common Applications-of-Integrals question type on ISC boards.

★ **Board Favorite** – This exact skeleton – solve for y or x , integrate between two given lines, first quadrant only – appears almost every year in some form (seen repeatedly across ISC 2018-2023 papers and every NCERT/Exemplar set for this chapter). Master this one type first.

Formula Used: $y^2 = kx \Rightarrow y = \sqrt{kx}$, $A = \int_{x_1}^{x_2} \sqrt{kx} \, dx$

Source: Exercise 8, Q1(i)



$y^2=x$ between $x=1$ and $x=4$, first quadrant strip shaded

Given: $y^2 = x$, $x = 1$, $x = 4$, x-axis, first quadrant

Solution:

Since $y^2 = x$ and we want the region in the first quadrant, take $y = \sqrt{x}$ (the positive branch, because the region asked for lies above the x-axis).

$$A = \int_1^4 \sqrt{x} \, dx$$

This integral sums up thin vertical strips of height $\sqrt{x}$ from $x = 1$ to $x = 4$.

$$A = \left[\frac{2}{3} x^{3/2} \right]_1^4 = \frac{2}{3} (4^{3/2} - 1^{3/2})$$

$4^{3/2} = 8$, so we substitute the upper and lower limits.

$$A = \frac{2}{3}(8 - 1) = \frac{2}{3} \times 7 = \frac{14}{3}$$

Final Answer

$$A = \frac{14}{3} \text{ sq. units}$$

Common Mistake: Forgetting that $y^2 = x$ gives $y = \pm\sqrt{x}$, and taking the wrong sign, or integrating w.r.t. the wrong variable when the curve opens sideways (it must be $\int y \, dx$ here, not $\int x \, dy$, since we're summing vertical strips under the curve).

Note – how this differs: Part (ii) below uses the identical method on a scaled parabola $y^2=9x$ – the only difference is the constant multiplying x inside the square root.

Sub-part (b): $y^2=9x$ form

Source: Exercise 8, Q1(ii)

Given: $y^2 = 9x$, $x = 2$, $x = 4$, x-axis, first quadrant

Solution:

Here $y = \sqrt{9x} = 3\sqrt{x}$.

$$A = \int_2^4 3\sqrt{x} \, dx = 3 \left[\frac{2}{3}x^{3/2} \right]_2^4 = 2 \left[x^{3/2} \right]_2^4$$

Substituting limits: $4^{3/2} = 8$ and $2^{3/2} = 2\sqrt{2}$.

$$A = 2(8 - 2\sqrt{2}) = 16 - 4\sqrt{2} = 4(4 - \sqrt{2})$$

Final Answer

$$A = 4(4 - \sqrt{2}) \text{ sq. units}$$

Note – how this differs: Part (c) swaps the roles of x and y : the parabola opens *upward* ($x^2=4y$), so the strip is cut by two horizontal lines, and we integrate x with respect to y instead.

Sub-part (c): $x^2=ky$ form (integrate w.r.t. y)

Source: Exercise 8, Q2(i)

Given: $x^2 = 4y$, $y = 2$, $y = 4$, y-axis, first quadrant

Solution:

Solve for x : $x = \sqrt{4y} = 2\sqrt{y}$. Since the bounds $y = 2$, $y = 4$ are horizontal lines, we now sum horizontal strips, i.e. integrate w.r.t. y .

$$A = \int_2^4 2\sqrt{y} \, dy = 2 \left[\frac{2}{3} y^{3/2} \right]_2^4 = \frac{4}{3} (4^{3/2} - 2^{3/2})$$

$$4^{3/2} = 8 \text{ and } 2^{3/2} = 2\sqrt{2}.$$

$$A = \frac{4}{3}(8 - 2\sqrt{2}) = \frac{32}{3} - \frac{8\sqrt{2}}{3} = \frac{8}{3}(4 - \sqrt{2})$$

Final Answer

$$A = \frac{8}{3}(4 - \sqrt{2}) \text{ sq. units}$$

Note – how this differs: Part (d) is a semicubical parabola (power $3/2$ instead of the usual square root) – the setup is otherwise identical to part (c).

Sub-part (d): Semicubical parabola $ay^2 = x^3$

Source: Exercise 8, Q2(ii)

Given: $ay^2 = x^3$, y -axis, $y = a$, $y = 2a$

Solution:

Solve for x in terms of y : $x^3 = ay^2 \Rightarrow x = a^{1/3}y^{2/3}$.

$$A = \int_a^{2a} a^{1/3} y^{2/3} \, dy = a^{1/3} \left[\frac{3}{5} y^{5/3} \right]_a^{2a}$$

We factor out $a^{1/3}$ since it does not depend on y , then substitute the limits.

$$A = \frac{3}{5} a^{1/3} [(2a)^{5/3} - a^{5/3}] = \frac{3}{5} a^{1/3} \cdot a^{5/3} [2^{5/3} - 1]$$

Combining the powers of a ($a^{1/3} \cdot a^{5/3} = a^2$):

$$A = \frac{3}{5} a^2 (2^{5/3} - 1)$$

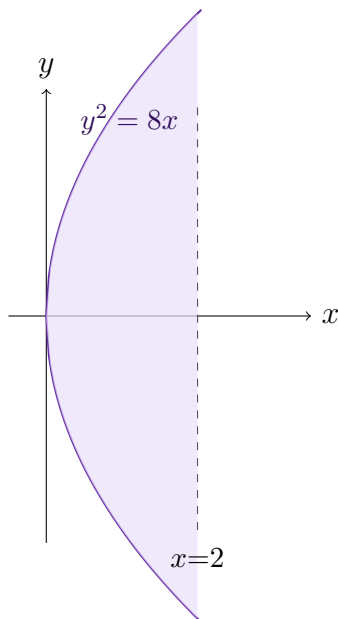
Final Answer

$$A = \frac{3}{5} a^2 (2^{5/3} - 1) \text{ sq. units}$$

Note – how this differs: Part (e) looks like part (a) but the region is NOT restricted to the first quadrant – the parabola's two branches (above and below the x -axis) both lie inside $x=2$, so the true region is symmetric about the x -axis and the first-quadrant strip must be doubled.

Sub-part (e): Region spans both quadrants – double the strip

Source: Exercise 8, Q3(i)



$y^2 = 8x$, region enclosed by the curve and $x=2$, both branches shaded

Given: $y^2 = 8x$, $x = 2$ (curve and line together enclose the region)

Solution:

The curve $y^2 = 8x$ has two symmetric branches, $y = \sqrt{8x}$ (above the x-axis) and $y = -\sqrt{8x}$ (below it). The line $x = 2$ cuts off a region that includes *both* branches, so we first find the first-quadrant (upper) area and then double it by symmetry.

$$A_{\text{upper}} = \int_0^2 \sqrt{8x} \, dx = 2\sqrt{2} \int_0^2 \sqrt{x} \, dx = 2\sqrt{2} \left[\frac{2}{3} x^{3/2} \right]_0^2$$

Substituting the limit ($2^{3/2} = 2\sqrt{2}$):

$$A_{\text{upper}} = 2\sqrt{2} \cdot \frac{2}{3} \cdot 2\sqrt{2} = \frac{2}{3} \times 8 = \frac{16}{3}$$

By symmetry about the x-axis, the lower branch contributes an equal area.

$$A = 2 \times A_{\text{upper}} = 2 \times \frac{16}{3} = \frac{32}{3}$$

Final Answer

$$A = \frac{32}{3} \text{ sq. units}$$

Common Mistake: Computing only the first-quadrant strip and forgetting to double it – always check whether the question restricts the region to "the first quadrant" (single branch) or not (both branches, must double).

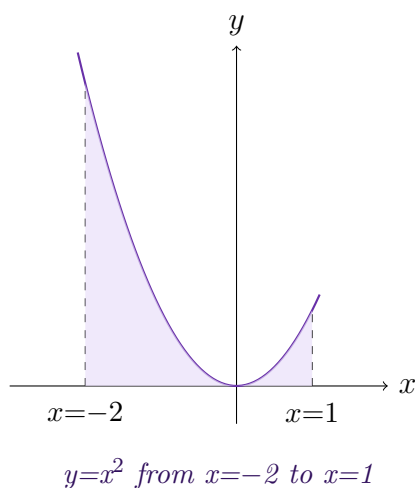
Quick Recall: Solve the parabola for whichever variable is being swept, integrate that expression between the two given bounds – and always check first whether the region is one branch (first quadrant only) or symmetric about the axis (double it).

Type 2: Area under $y=f(x)$, a general polynomial curve

The boundary is given directly as y in terms of x (not a standard conic), with the region cut off between two vertical ordinates and the x -axis. The method is the same basic vertical-strip integral, but always check the sign of y across the interval before integrating – if the curve dips below the x -axis anywhere in the interval, that portion must be treated separately (see Type 3's alternate example, which needs exactly this care).

Formula Used: $A = \int_{x_1}^{x_2} y \, dx$

Source: Exercise 8, Q3(ii)



Given: $y = x^2$, $x = -2$, $x = 1$, x -axis

Solution:

Since $y = x^2 \geq 0$ for every real x , the curve never dips below the x -axis on $[-2, 1]$, so a single integral covers the whole region – no splitting needed here.

$$A = \int_{-2}^1 x^2 \, dx = \left[\frac{x^3}{3} \right]_{-2}^1$$

Substitute the upper and lower limits.

$$A = \frac{1^3}{3} - \frac{(-2)^3}{3} = \frac{1}{3} - \left(\frac{-8}{3} \right) = \frac{1}{3} + \frac{8}{3} = \frac{9}{3}$$

$$A = 3$$

Final Answer

$A = 3$ sq. units

Common Mistake: Assuming every polynomial region needs the modulus/splitting treatment – check the sign of y across the whole interval first; if it never changes sign, one plain integral suffices.

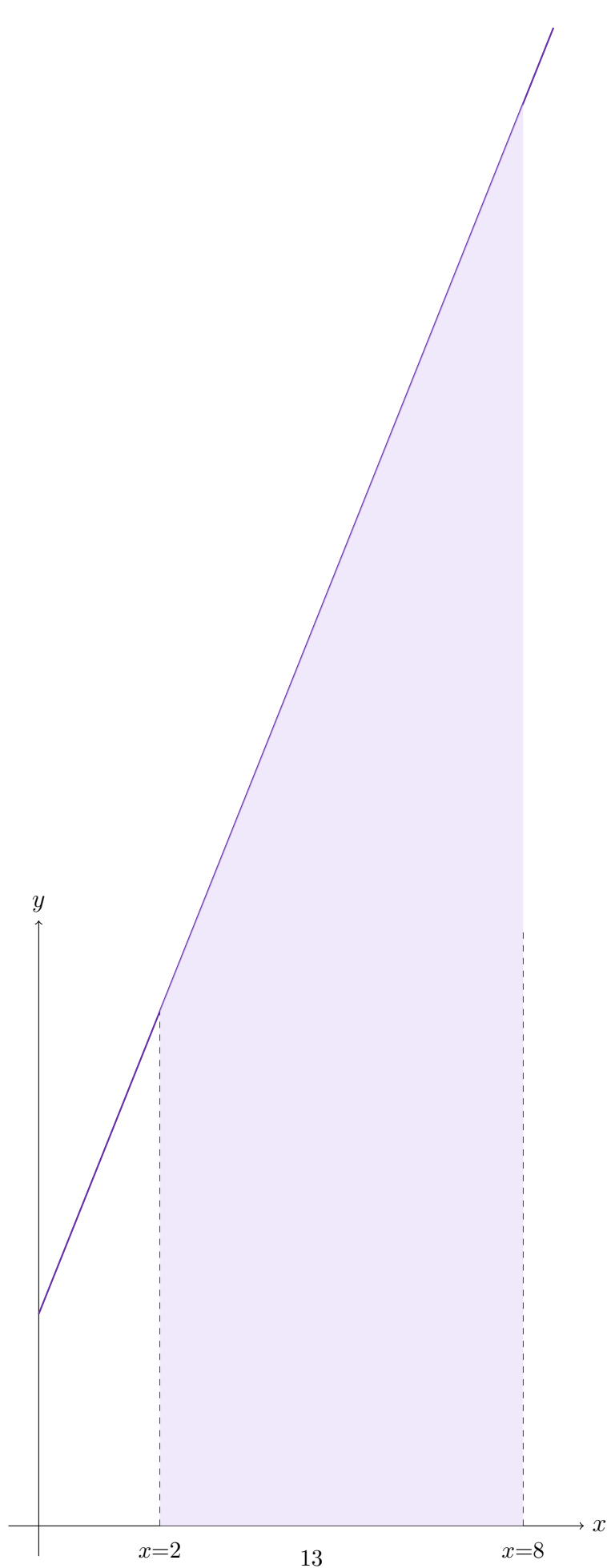
Quick Recall: Plug in $y = f(x)$ directly and integrate – but always scan the interval for a sign change in y before you integrate in one shot.

Type 3: Area under a straight line (trapezium via integration)

The boundary is a straight line and the region is cut off between two vertical ordinates and the x-axis – geometrically a trapezium, but the question specifically asks you to find it "using integration". Solve the line for y and integrate as usual; if the line crosses the x-axis inside the given interval, split the integral at the crossing point and add the absolute values of each piece (a trapezium-shaped question can quietly turn into two triangles).

Formula Used: $y = mx + c$, $A = \int_{x_1}^{x_2} (mx + c) dx$

Source: Exercise 8, Q4



Line $2y=5x+7$ between $x=2$ and $x=8$

Given: $2y = 5x + 7$, $x = 2$, $x = 8$, x-axis

Solution:

Rewrite the line as $y = \frac{5x+7}{2}$. On $[2, 8]$ this is always positive, so one integral suffices.

$$A = \int_2^8 \frac{5x+7}{2} dx = \frac{1}{2} \left[\frac{5x^2}{2} + 7x \right]_2^8$$

Substitute $x = 8$ and $x = 2$ in turn.

$$\begin{aligned} &= \frac{1}{2} \left[\left(\frac{5(64)}{2} + 56 \right) - \left(\frac{5(4)}{2} + 14 \right) \right] = \frac{1}{2} [(160 + 56) - (10 + 14)] \\ &= \frac{1}{2} [216 - 24] = \frac{1}{2} (192) = 96 \end{aligned}$$

Final Answer

$A = 96$ sq. units

Common Mistake: Not checking whether the line changes sign inside the interval – if it does (see the alternate example below), integrating across the crossing in one shot silently cancels area instead of adding it.

Alternate Example – line that crosses the x-axis (needs splitting)

Source: Exercise 8, Q34

Given: $y = 3x + 2$, $x = -2$, $x = 1$, x-axis

Solution:

First find where the line crosses the x-axis: set $3x + 2 = 0 \Rightarrow x = -\frac{2}{3}$. This point lies inside $[-2, 1]$, so the line is below the x-axis on $[-2, -\frac{2}{3}]$ and above it on $[-\frac{2}{3}, 1]$ – the two pieces must be handled separately.

Case: $-2 \leq x \leq -\frac{2}{3}$ (line below the axis)

$$\int_{-2}^{-2/3} (3x+2) dx = \left[\frac{3x^2}{2} + 2x \right]_{-2}^{-2/3} = \left(\frac{2}{3} - \frac{4}{3} \right) - (6 - 4) = -\frac{2}{3} - 2 = -\frac{8}{3}$$

This integral is negative because the curve lies below the x-axis here; the actual area is its absolute value, $\frac{8}{3}$.

Case: $-\frac{2}{3} \leq x \leq 1$ (line above the axis)

$$\int_{-2/3}^1 (3x+2) dx = \left[\frac{3x^2}{2} + 2x \right]_{-2/3}^1 = \left(\frac{3}{2} + 2 \right) - \left(-\frac{2}{3} \right) = \frac{7}{2} + \frac{2}{3} = \frac{25}{6}$$

This piece is already positive, so it is added as-is.

$$A = \left| -\frac{8}{3} \right| + \frac{25}{6} = \frac{16}{6} + \frac{25}{6} = \frac{41}{6}$$

Final Answer

$$A = \frac{41}{6} \text{ sq. units}$$

Quick Recall: A straight-line area is just a trapezium formula in disguise – but always find where the line crosses the x-axis first, and split there if the crossing lies inside your interval.

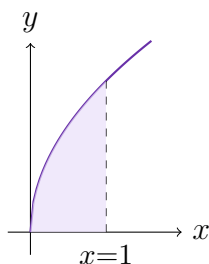
Type 4: Area under $y=k\sqrt{x}$ (or $\sqrt{\text{linear}}$)

The curve is a square-root function – either $y = k\sqrt{x}$ directly, or $y = \sqrt{x - a}$ shifted along the x-axis. The method is a plain vertical-strip integral; the only extra care needed is remembering the power rule works with $\sqrt{x} = x^{1/2}$, so $\int \sqrt{x} \, dx = \frac{2}{3}x^{3/2}$.

★ **Board Favorite** – Square-root curves are a favourite because they combine a "sketch the curve" instruction with a clean fractional-power integral – a very standard Exemplar-style pairing.

Formula Used: $A = \int_{x_1}^{x_2} k\sqrt{x} \, dx$

Source: Exercise 8, Q5



$y=2\sqrt{x}$ between $x=0$ and $x=1$

Given: $y = 2\sqrt{x}$, $x = 0$, $x = 1$

Solution:

$$A = \int_0^1 2\sqrt{x} \, dx = 2 \left[\frac{2}{3}x^{3/2} \right]_0^1 = \frac{4}{3} (1^{3/2} - 0)$$

$$A = \frac{4}{3}$$

Final Answer

$$A = \frac{4}{3} \text{ sq. units}$$

Common Mistake: Writing the power rule wrong for a fractional exponent – $\int x^{1/2} dx = \frac{2}{3}x^{3/2}$, not $\frac{1}{2}x^{-1/2}$ (that would be the derivative, not the integral).

Alternate Board-Style Example

Source: Exercise 8, Q6

Given: $y = \sqrt{x-1}$, $x = 1$, $x = 5$

Solution:

This is the same curve shape as $y = \sqrt{x}$, shifted right by 1 unit – so substitute $u = x - 1$ mentally, or integrate directly with the shifted variable.

$$A = \int_1^5 \sqrt{x-1} \, dx = \left[\frac{2}{3}(x-1)^{3/2} \right]_1^5$$

At $x = 5$: $(x-1)^{3/2} = 4^{3/2} = 8$. At $x = 1$: $(x-1)^{3/2} = 0$.

$$A = \frac{2}{3}(8 - 0) = \frac{16}{3}$$

Final Answer

$$A = \frac{16}{3} \text{ sq. units}$$

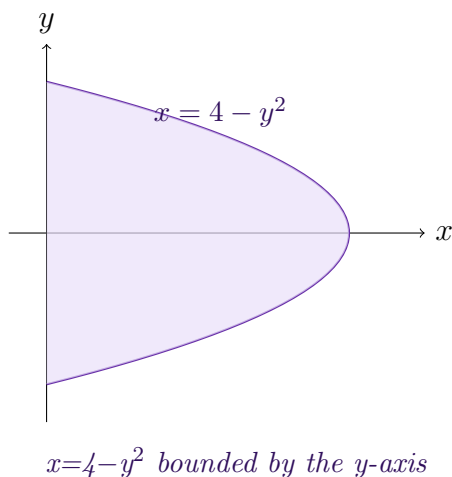
Quick Recall: Treat $\sqrt{x}$ and $\sqrt{x-a}$ the same way – power rule with exponent $3/2$, shifted bounds if the square root has $(x-a)$ inside it.

Type 5: Area bounded by a sideways-left parabola and the y-axis

The curve is written as x in terms of y with a negative y^2 term (opens toward the left), and the region is bounded by the curve and the y-axis itself (not by two separate lines). Find where the curve meets the y-axis first (that gives the y -limits), then integrate x w.r.t. y ; the region is typically symmetric about the x-axis, so integrate the top half and double it.

Formula Used: $x = c - y^2$, $A = \int_{-y_0}^{y_0} (c - y^2) \, dy$

Source: Exercise 8, Q7



Given: $x = 4 - y^2$, y-axis ($x = 0$)

Solution:

First find where the curve meets the y-axis: set $x = 0$.

$$4 - y^2 = 0 \Rightarrow y^2 = 4 \Rightarrow y = \pm 2$$

So the region runs from $y = -2$ to $y = 2$. Since the parabola is symmetric about the x-axis, integrate the upper half ($y = 0$ to 2) and double it.

$$A = 2 \int_0^2 (4 - y^2) dy = 2 \left[4y - \frac{y^3}{3} \right]_0^2$$

Substitute $y = 2$:

$$A = 2 \left[8 - \frac{8}{3} \right] = 2 \times \frac{16}{3} = \frac{32}{3}$$

Final Answer

$$A = \frac{32}{3} \text{ sq. units}$$

Common Mistake: Integrating $x = 4 - y^2$ with respect to x instead of y – because the curve is given as $x = f(y)$, the strips here are horizontal, so the integral must be $\int x dy$.

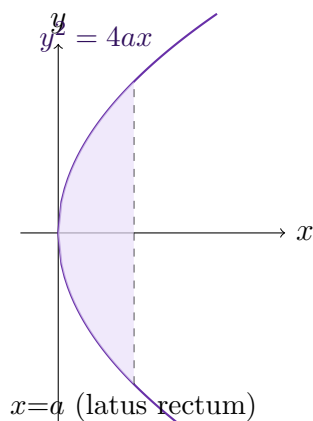
Quick Recall: Find where the curve crosses the y-axis to get the y -limits, then integrate $x = f(y)$ directly with respect to y – double the half if the region is symmetric.

Type 6: Area of a parabola cut off by its own latus rectum

A parabola in the standard form $y^2 = 4ax$, and the "line" cutting it off is not given explicitly but is the latus rectum itself, i.e. the vertical line through the focus, $x = a$. Recognising this is the whole trick – once you know $x = a$ is the bound, the integral is identical to Type 1.

Formula Used: $y^2 = 4ax$ has latus rectum $x = a$, $A = 2 \int_0^a \sqrt{4ax} \, dx$

Source: Exercise 8, Q8



$y^2 = 4ax$ cut by its latus rectum $x = a$

Given: $y^2 = 4ax$

Solution:

The latus rectum of $y^2 = 4ax$ is the vertical chord through the focus $(a, 0)$, i.e. the line $x = a$. This is not stated explicitly in the question – recognising it is the key step.

$$A = 2 \int_0^a \sqrt{4ax} \, dx = 2 \times 2\sqrt{a} \int_0^a \sqrt{x} \, dx = 4\sqrt{a} \left[\frac{2}{3} x^{3/2} \right]_0^a$$

Substitute $x = a$ ($a^{3/2} = a\sqrt{a}$):

$$A = 4\sqrt{a} \times \frac{2}{3} \times a\sqrt{a} = \frac{8}{3}a^2$$

Final Answer

$$A = \frac{8}{3}a^2 \text{ sq. units}$$

Common Mistake: Not recognising "latus rectum" as $x = a$ and leaving the answer in terms of an undefined bound, or confusing the latus rectum with the directrix ($x = -a$, which is outside the parabola).

Quick Recall: "Latus rectum of $y^2 = 4ax$ " always means the vertical line $x = a$ – once you spot that translation, it's a standard double-integral like Type 1.

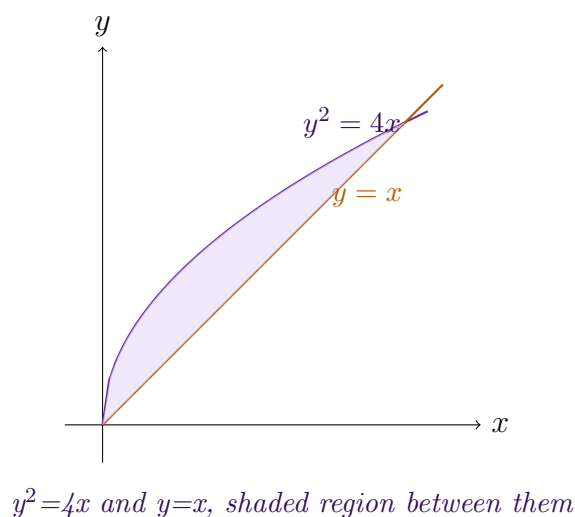
II. Area Between Two Curves

Type 7: Area between a parabola and a chord through the origin

A parabola in standard form ($y^2=4ax$) and a line through the origin ($y=mx$) intersect at exactly two points – the origin and one other point. Find that second intersection point first, then integrate (curve – line) between the two x-limits, checking which one is on top.

Formula Used: $y^2 = 4ax$ and $y = mx$ meet at $x = 0$ and $x = \frac{4a}{m^2}$

Source: Exercise 8, Q9(i)



Given: $y^2 = 4x$, $y = x$

Solution:

Find the intersection points by substitution. From $y = x$, put $x = y$ into $y^2 = 4x$:

$$y^2 = 4y \Rightarrow y^2 - 4y = 0 \Rightarrow y(y - 4) = 0 \Rightarrow y = 0, 4$$

So the curves meet at $(0, 0)$ and $(4, 4)$. On $(0, 4)$, check which curve is on top – at $x = 1$, the parabola gives $y = 2$ while the line gives $y = 1$, so the parabola is the upper boundary.

$$A = \int_0^4 (2\sqrt{x} - x) dx = \left[2 \cdot \frac{2}{3} x^{3/2} - \frac{x^2}{2} \right]_0^4$$

Substitute $x = 4$ ($4^{3/2} = 8$):

$$A = \frac{4}{3}(8) - \frac{16}{2} = \frac{32}{3} - 8 = \frac{8}{3}$$

Final Answer

$$A = \frac{8}{3} \text{ sq. units}$$

Common Mistake: Subtracting (line – curve) instead of (curve – line) – always test a point strictly between the two intersection x-values to see which boundary is actually on top.

Alternate Board-Style Example – general parabola and chord

Source: Exercise 8, Q9(ii)

Given: $y^2 = 4ax$, $y = mx$ ($m > 0$)

Solution:

Substitute $x = y/m$ into $y^2 = 4ax$:

$$y^2 = \frac{4ay}{m} \Rightarrow y \left(y - \frac{4a}{m} \right) = 0 \Rightarrow y = 0, \frac{4a}{m}$$

Converting to x : the second intersection is at $x = \frac{4a}{m^2}$. As before, the parabola lies above the chord between the two intersections.

$$A = \int_0^{4a/m^2} \left(\sqrt{4ax} - mx \right) dx = \left[2\sqrt{a} \cdot \frac{2}{3} x^{3/2} - \frac{mx^2}{2} \right]_0^{4a/m^2}$$

Substituting $x = \frac{4a}{m^2}$ carefully (using $x^{3/2} = \frac{8a^{3/2}}{m^3}$ and $x^2 = \frac{16a^2}{m^4}$):

$$A = \frac{4}{3} \sqrt{a} \cdot \frac{8a^{3/2}}{m^3} - \frac{m}{2} \cdot \frac{16a^2}{m^4} = \frac{32a^2}{3m^3} - \frac{8a^2}{m^3}$$

$$A = \frac{32a^2}{3m^3} - \frac{24a^2}{3m^3} = \frac{8a^2}{3m^3}$$

Final Answer

$$A = \frac{8a^2}{3m^3} \text{ sq. units}$$

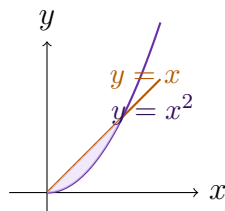
Quick Recall: Substitute the line into the parabola to get the second intersection point, confirm which curve is on top with a test value, then integrate (top – bottom).

Type 8: Area between $y=x$ and $y=x^2$

The simplest "region between two curves" question – a straight line and a parabola crossing twice. Same method as Type 7 but with both curves given directly, no chord/parabola disguise.

Formula Used: $A = \int_{x_1}^{x_2} [y_{\text{upper}} - y_{\text{lower}}] dx$

Source: Exercise 8, Q10



$y=x$ and $y=x^2$, shaded lens-shaped region

Given: $y = x$, $y = x^2$

Solution:

Find intersections: $x = x^2 \Rightarrow x^2 - x = 0 \Rightarrow x(x - 1) = 0 \Rightarrow x = 0, 1$. Between $x = 0$ and $x = 1$, test $x = 0.5$: line gives 0.5, parabola gives 0.25 – so the line is on top.

$$A = \int_0^1 (x - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

Final Answer

$$A = \frac{1}{6} \text{ sq. units}$$

Common Mistake: Assuming the parabola is always on top just because it's the "curvier" boundary – always test a point.

Quick Recall: Set the two expressions equal to find intersections, test the midpoint to see which is on top, then integrate the difference.

Type 9: Area of a polygonal region using integration

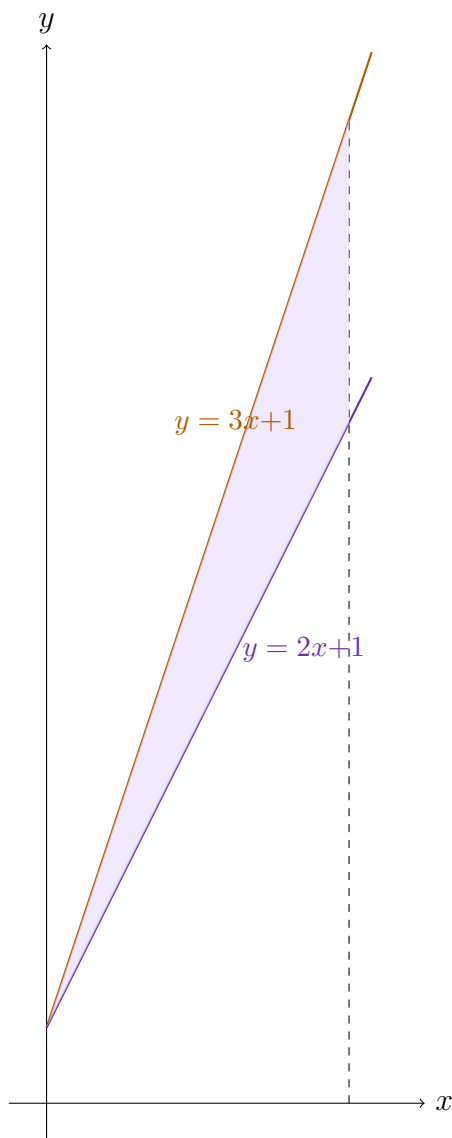
The region is a triangle (or other polygon), but the question insists you use integration rather than the coordinate-geometry area formula. Two flavours appear: (a) the three side-equations are given directly, or (b) only the vertices are given and you must first find the three side-equations yourself. Either way, once you have the equations, split the x-range at the "middle" vertex and integrate (upper line – lower line) on each piece.

★ **Board Favorite** – Triangular-region-by-integration is one of the most frequently repeated Long Answer questions in ISC Maths – it tests both coordinate geometry (finding line equations) and integration in one question, which examiners like.

Formula Used: Line through two points: $\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$

Formula Used: $A = \int_{x_1}^{x_m} [y_{\text{top}} - y_{\text{bottom}}] dx + \int_{x_m}^{x_2} [y_{\text{top}} - y_{\text{bottom}}] dx$

Source: Exercise 8, Q11



Triangle formed by $y=2x+1$, $y=3x+1$, and $x=4$

Given: $y = 2x + 1$, $y = 3x + 1$, $x = 4$

Solution:

Both lines pass through $(0, 1)$ – setting $2x + 1 = 3x + 1$ gives $x = 0$. So the triangle has one vertex at $(0, 1)$ and the opposite side is the vertical line $x = 4$; no splitting is needed since a single vertex sits at the left end. Test $x = 1$: line $y = 3x + 1$ gives 4, line

$y = 2x + 1$ gives 3, so $y = 3x + 1$ is the upper boundary throughout.

$$A = \int_0^4 [(3x + 1) - (2x + 1)] dx = \int_0^4 x dx = \left[\frac{x^2}{2} \right]_0^4 = 8$$

Final Answer

$A = 8$ sq. units

Common Mistake: Not checking whether the triangle's vertex lies inside the integration interval – if it does, a single top-minus-bottom integral is wrong and the range must be split there.

Note – how this differs: Part (b) below has all three sides genuinely cutting across the region, so the x-range must be split at the middle vertex – unlike Q11 above, where one vertex conveniently sat at the edge.

Sub-part: three general lines, needs splitting

Source: Exercise 8, Q13(iii)

Given: $4x - y + 5 = 0$, $x + y - 5 = 0$, $x - 4y + 5 = 0$

Solution:

Rewrite each as $y = \dots$: $L_1 : y = 4x + 5$, $L_2 : y = 5 - x$, $L_3 : y = \frac{x + 5}{4}$.

Find the three vertices by solving pairs of equations.

$$L_1 = L_2 : 4x + 5 = 5 - x \Rightarrow x = 0, y = 5 \Rightarrow (0, 5)$$

$$L_2 = L_3 : 5 - x = \frac{x + 5}{4} \Rightarrow 20 - 4x = x + 5 \Rightarrow x = 3, y = 2 \Rightarrow (3, 2)$$

$$L_1 = L_3 : 4x + 5 = \frac{x + 5}{4} \Rightarrow 16x + 20 = x + 5 \Rightarrow x = -1, y = 1 \Rightarrow (-1, 1)$$

The vertices are $(-1, 1)$, $(0, 5)$, $(3, 2)$ – so the "middle" x-value is $x = 0$, and we split there.

Case: $-1 \leq x \leq 0$ (top: L_1 , bottom: L_3)

$$\int_{-1}^0 \left[(4x + 5) - \frac{x + 5}{4} \right] dx = \int_{-1}^0 \frac{15x + 15}{4} dx = \left[\frac{15x^2}{8} + \frac{15x}{4} \right]_{-1}^0 = 0 - \left(\frac{15}{8} - \frac{15}{4} \right) = \frac{15}{8}$$

Case: $0 \leq x \leq 3$ (top: L_2 , bottom: L_3)

$$\int_0^3 \left[(5 - x) - \frac{x + 5}{4} \right] dx = \int_0^3 \frac{15 - 5x}{4} dx = \left[\frac{15x}{4} - \frac{5x^2}{8} \right]_0^3 = \frac{45}{4} - \frac{45}{8} = \frac{45}{8}$$

$$A = \frac{15}{8} + \frac{45}{8} = \frac{60}{8} = \frac{15}{2}$$

Final Answer

$$A = \frac{15}{2} \text{ sq. units}$$

Common Mistake: Forgetting to split at the middle vertex, or splitting at the wrong x-value – always solve all three pairs of equations first to find all three vertices before deciding where to split.

Note – how this differs: Sub-part (c) is the vertices-only flavour: no line equations are given, so the extra first step is finding all three side-equations from the coordinates before the integration even starts. Note this happens to be the very same triangle as the alternate example above.

Sub-part: vertices given – find side equations first

Source: Exercise 8, Q12(i)

Given: Vertices $(-1, 1)$, $(0, 5)$, $(3, 2)$

Solution:

Use the two-point line formula for each side. Side $(-1, 1)$ – $(0, 5)$: slope $= \frac{5-1}{0-(-1)} = 4$, so $y-1 = 4(x+1) \Rightarrow y = 4x+5$. Side $(0, 5)$ – $(3, 2)$: slope $= \frac{2-5}{3-0} = -1$, so $y = 5-x$. Side $(-1, 1)$ – $(3, 2)$: slope $= \frac{2-1}{3-(-1)} = \frac{1}{4}$, so $y = \frac{x+5}{4}$. These are exactly L_1, L_2, L_3 from the example above, so the integration proceeds identically (split at $x = 0$), giving the same result.

Final Answer

$$A = \frac{15}{2} \text{ sq. units}$$

Common Mistake: Getting a side's slope backwards (swapping which point is "first") – always keep the same point order in numerator and denominator of the slope formula.

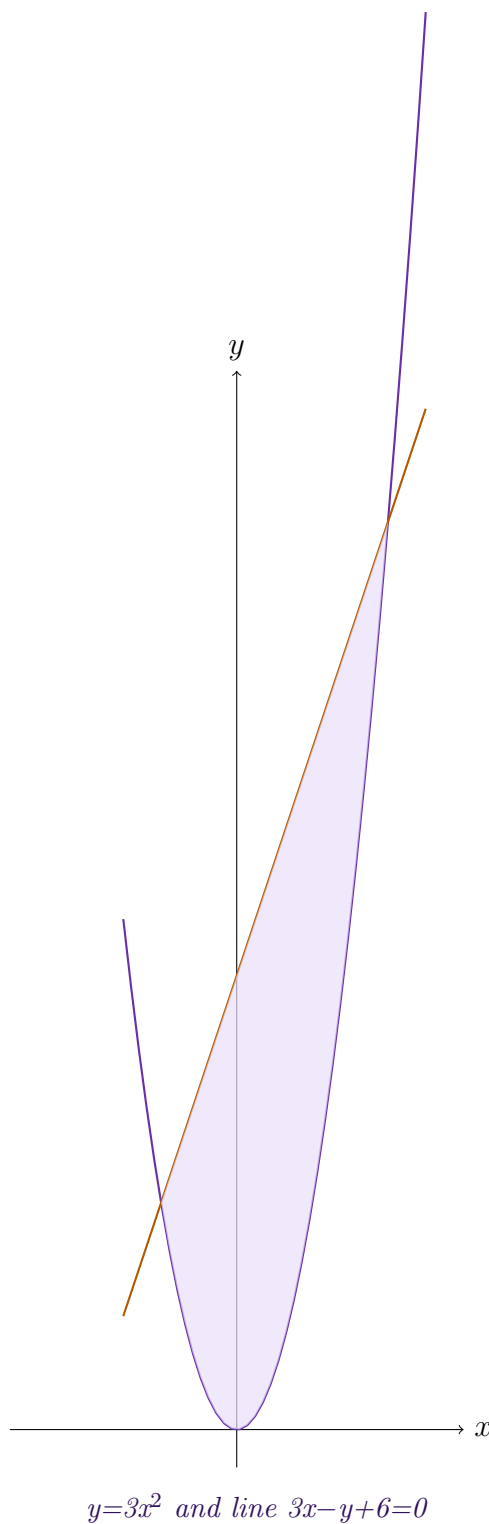
Quick Recall: Find all vertices (solving line pairs, or forming lines from given points), identify the middle vertex's x-coordinate, split the integral there, and integrate (top line – bottom line) on each piece.

Type 10: Area between a parabola and a line – mixed forms

The general "parabola meets line" question, but the parabola can be given in any orientation ($y=kx^2$, $x=\sqrt{y}$, $x^2=ky$, $y^2=kx$) and the boundary can include an axis. Always find the intersection points first; if a boundary (like the y-axis) only applies over part of the range, split the integral there.

Formula Used: $A = \int [y_{\text{line}} - y_{\text{curve}}] dx$ (or the y-integral analogue)

Source: Exercise 8, Q23(ii)



Given: $y = 3x^2$, $3x - y + 6 = 0$ (i.e. $y = 3x + 6$)

Solution:

Set the two expressions for y equal.

$$3x^2 = 3x + 6 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0 \Rightarrow x = -1, 2$$

Test $x = 0$: line gives 6, parabola gives 0 – the line is on top throughout.

$$A = \int_{-1}^2 [(3x + 6) - 3x^2] dx = \left[\frac{3x^2}{2} + 6x - x^3 \right]_{-1}^2$$

At $x = 2$: $6 + 12 - 8 = 10$. At $x = -1$: $1.5 - 6 + 1 = -3.5$.

$$A = 10 - (-3.5) = 13.5 = \frac{27}{2}$$

Final Answer

$$A = \frac{27}{2} \text{ sq. units}$$

Note – how this differs: Sub-part (b) swaps roles: the curve is written as $x = \sqrt{y}$ and one boundary is the y-axis, which is only the true left edge for part of the range – the integral must be split where the line crosses over to become the tighter boundary.

Sub-part: curve $x = \sqrt{y}$, split at handover point

Source: Exercise 8, Q23(iii)

Given: $x - y + 2 = 0$ (i.e. $x = y - 2$), $x = \sqrt{y}$, y-axis

Solution:

Find where the line and curve meet: $y - 2 = \sqrt{y}$. Squaring, $y^2 - 4y + 4 = y \Rightarrow y^2 - 5y + 4 = 0 \Rightarrow (y - 1)(y - 4) = 0 \Rightarrow y = 1, 4$. Checking in the original (unsquared) equation, only $y = 4$ satisfies $y - 2 = \sqrt{y}$ (since $y = 1$ gives $-1 \neq 1$), so the curves genuinely meet at $y = 4$. For $0 \leq y \leq 2$ the line $x = y - 2$ is negative, so the y-axis ($x = 0$) is the true left boundary there instead.

Case: $0 \leq y \leq 2$ (left: y-axis, right: curve)

$$\int_0^2 \sqrt{y} dy = \left[\frac{2}{3} y^{3/2} \right]_0^2 = \frac{2}{3} (2\sqrt{2}) = \frac{4\sqrt{2}}{3}$$

Case: $2 \leq y \leq 4$ (left: line, right: curve)

$$\int_2^4 [\sqrt{y} - (y - 2)] dy = \left[\frac{2}{3} y^{3/2} - \frac{y^2}{2} + 2y \right]_2^4$$

At $y = 4$: $\frac{16}{3} - 8 + 8 = \frac{16}{3}$. At $y = 2$: $\frac{4\sqrt{2}}{3} - 2 + 4 = \frac{4\sqrt{2}}{3} + 2$.

$$= \frac{16}{3} - \frac{4\sqrt{2}}{3} - 2 = \frac{10 - 4\sqrt{2}}{3}$$

$$A = \frac{4\sqrt{2}}{3} + \frac{10 - 4\sqrt{2}}{3} = \frac{10}{3}$$

Final Answer

$$A = \frac{10}{3} \text{ sq. units}$$

Common Mistake: Squaring $y - 2 = \sqrt{y}$ and accepting both roots without checking back in the original equation – squaring can introduce a spurious root ($y = 1$ here), which must be discarded.

Note – how this differs: Sub-part (c) uses $x^2 = 4y$ (upward parabola) with a shifted line, same method as (a) but with roles of x and y matching the parabola's orientation.

Sub-part: $x^2 = 4y$ and a line

Source: Exercise 8, Q24(i)

Given: $x^2 = 4y$ (i.e. $y = x^2/4$), $x = 4y - 2$ (i.e. $y = (x + 2)/4$)

Solution:

Equate: $\frac{x^2}{4} = \frac{x + 2}{4} \Rightarrow x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0 \Rightarrow x = -1, 2$.

Testing $x = 0$: line gives 0.5, curve gives 0 – line on top.

$$A = \int_{-1}^2 \left[\frac{x+2}{4} - \frac{x^2}{4} \right] dx = \frac{1}{4} \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2$$

At $x = 2$: $2 + 4 - \frac{8}{3} = \frac{10}{3}$. At $x = -1$: $0.5 - 2 + \frac{1}{3} = -\frac{7}{6}$.

$$A = \frac{1}{4} \left(\frac{10}{3} + \frac{7}{6} \right) = \frac{1}{4} \times \frac{27}{6} = \frac{9}{8}$$

Final Answer

$$A = \frac{9}{8} \text{ sq. units}$$

Note – how this differs: Sub-part (d) is a region given by two *inequalities* instead of a boundary description – $y^2 \leq 2x$ means "inside/right of the parabola" and $y \geq 4x - 1$ means "above the line". Translating inequalities into a picture is the extra skill tested here.

Sub-part: region from inequalities

Source: Exercise 8, Q24(iii)

Given: $(x, y) : y^2 \leq 2x, \quad y \geq 4x - 1$

Solution:

$y^2 \leq 2x$ means $x \geq \frac{y^2}{2}$ (right of the parabola $x = y^2/2$). $y \geq 4x - 1$ means $x \leq \frac{y+1}{4}$ (left of the line $x = (y+1)/4$). So for each y , x runs from the parabola to the line. Find where they cross:

$$\frac{y^2}{2} = \frac{y+1}{4} \Rightarrow 2y^2 = y+1 \Rightarrow 2y^2 - y - 1 = 0 \Rightarrow (2y+1)(y-1) = 0 \Rightarrow y = -\frac{1}{2}, 1$$

$$A = \int_{-1/2}^1 \left[\frac{y+1}{4} - \frac{y^2}{2} \right] dy = \left[\frac{y^2}{8} + \frac{y}{4} - \frac{y^3}{6} \right]_{-1/2}^1$$

At $y = 1$: $\frac{1}{8} + \frac{1}{4} - \frac{1}{6} = \frac{3}{24} + \frac{6}{24} - \frac{4}{24} = \frac{5}{24}$. At $y = -\frac{1}{2}$: $\frac{1}{32} - \frac{1}{8} + \frac{1}{48}$; putting over 96: $\frac{3}{96} - \frac{12}{96} + \frac{2}{96} = -\frac{7}{96}$.

$$A = \frac{5}{24} - \left(-\frac{7}{96} \right) = \frac{20}{96} + \frac{7}{96} = \frac{27}{96} = \frac{9}{32}$$

Final Answer

$$A = \frac{9}{32} \text{ sq. units}$$

Common Mistake: Getting the inequality direction backwards – always pick a simple test point (e.g. the origin) and check whether it satisfies the inequality, to be sure which side is being described.

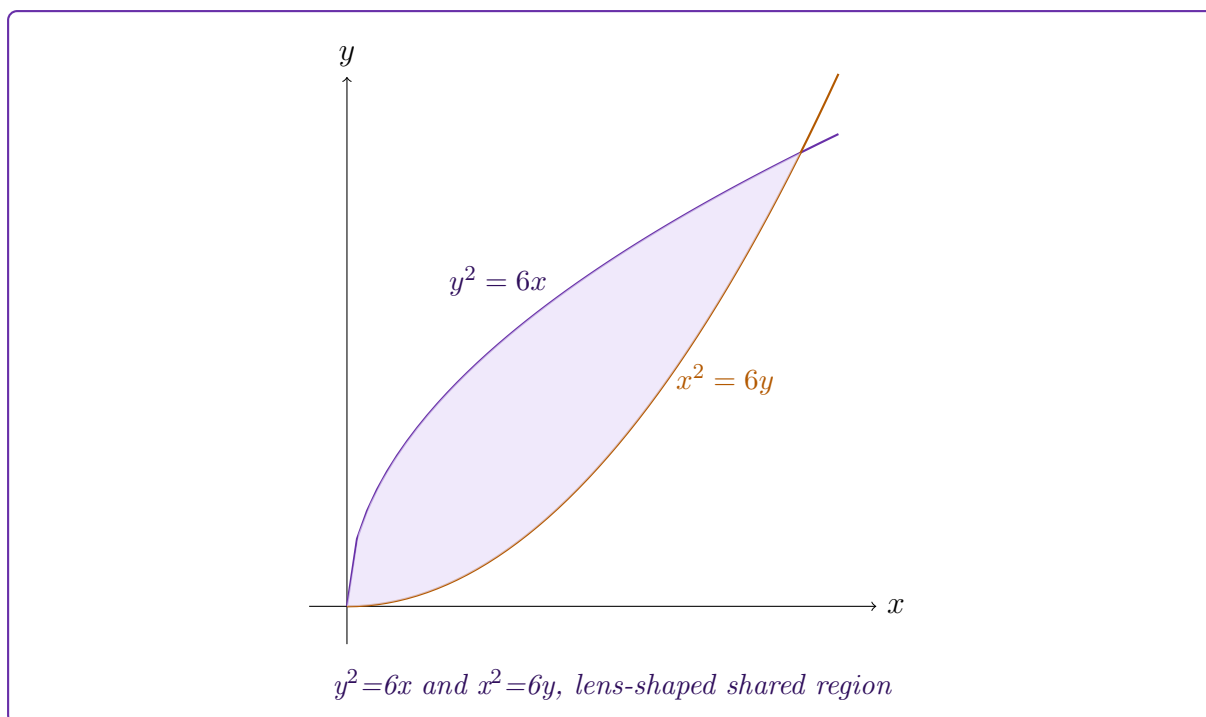
Quick Recall: Whatever orientation the parabola is in, equate the two expressions to find the x- or y-limits, test a point to see which is on top, and watch for a boundary (axis) that only applies to part of the range.

Type 11: Area between two parabolas

Two parabolas symmetric about the line $y=x$ (one opens along each axis) always intersect at the origin and exactly one other point. Solve the pair simultaneously (substitute one into the other), integrate the difference, and the algebra typically collapses very cleanly because of the symmetry.

Formula Used: $y^2 = kx$ and $x^2 = ky$ meet at $(0,0)$ and (k,k)

Source: Exercise 8, Q25(i)



Given: $y^2 = 6x$, $x^2 = 6y$

Solution:

From $y^2 = 6x$, $x = \frac{y^2}{6}$. Substitute into $x^2 = 6y$:

$$\left(\frac{y^2}{6}\right)^2 = 6y \Rightarrow \frac{y^4}{36} = 6y \Rightarrow y^4 = 216y \Rightarrow y(y^3 - 216) = 0$$

So $y = 0$ or $y^3 = 216 \Rightarrow y = 6$; correspondingly $x = 0$ or $x = 6$. The curves meet at $(0, 0)$ and $(6, 6)$. Testing $x = 1$: $y^2 = 6x$ gives $y = \sqrt{6} \approx 2.45$, while $x^2 = 6y$ gives $y = 1/6 \approx 0.17$ – so $y^2 = 6x$ is the upper curve.

$$A = \int_0^6 \left[\sqrt{6x} - \frac{x^2}{6} \right] dx = \left[\sqrt{6} \cdot \frac{2}{3} x^{3/2} - \frac{x^3}{18} \right]_0^6$$

At $x = 6$ ($6^{3/2} = 6\sqrt{6}$): $\sqrt{6} \times \frac{2}{3} \times 6\sqrt{6} - \frac{216}{18} = \frac{2}{3} \times 36 - 12 = 24 - 12$.

$$A = 12$$

Final Answer

$A = 12$ sq. units

Common Mistake: Forgetting the origin is always one intersection point of two parabolas through it, and missing the trivial $y = 0$ root when solving the quartic.

Alternate Board-Style Example

Source: Exercise 8, Q25(ii)

Given: $y^2 = 2px, \quad x^2 = 2py$

Solution:

By the same method, from $x = \frac{y^2}{2p}$ substituted into $x^2 = 2py$:

$$\frac{y^4}{4p^2} = 2py \Rightarrow y^4 = 8p^3y \Rightarrow y = 0 \text{ or } y = 2p$$

The curves meet at $(0, 0)$ and $(2p, 2p)$.

$$A = \int_0^{2p} \left[\sqrt{2px} - \frac{x^2}{2p} \right] dx = \left[\sqrt{2p} \cdot \frac{2}{3} x^{3/2} - \frac{x^3}{6p} \right]_0^{2p}$$

At $x = 2p$ (using $(2p)^{3/2} = 2p\sqrt{2p}$):

$$\sqrt{2p} \times \frac{2}{3} \times 2p\sqrt{2p} - \frac{8p^3}{6p} = \frac{2}{3} \times 4p^2 - \frac{4p^2}{3} = \frac{8p^2}{3} - \frac{4p^2}{3}$$

$$A = \frac{4p^2}{3}$$

Final Answer

$$A = \frac{4}{3}p^2 \text{ sq. units}$$

Quick Recall: Two parabolas opening along different axes always meet at the origin and one other symmetric point – substitute one equation into the other, solve, then integrate the (upper – lower) difference.

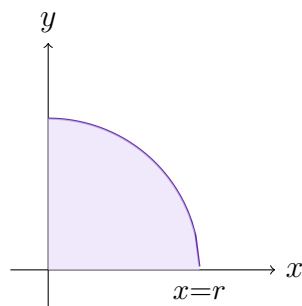
III. Circles and Ellipses

Type 12: Area under / enclosed by a semicircle

The curve is $y = \sqrt{r^2 - x^2}$, the upper half of a circle. "Area under the curve between $x = 0$ and $x = r$ " gives one quadrant; multiplying by symmetry gives the semicircle or full circle. This type is also used to *derive* the circle-area formula from first principles – a classic "prove that" question.

Formula Used: $\int \sqrt{r^2 - x^2} dx = \frac{x}{2} \sqrt{r^2 - x^2} + \frac{r^2}{2} \sin^{-1}\left(\frac{x}{r}\right) + C$

Source: Exercise 8, Q14



Quarter circle $y = \sqrt{r^2 - x^2}$, x from 0 to r

Given: $y = \sqrt{r^2 - x^2}$, $x = 0$, $x = r$

Solution:

$$A = \int_0^r \sqrt{r^2 - x^2} dx = \left[\frac{x}{2} \sqrt{r^2 - x^2} + \frac{r^2}{2} \sin^{-1}\left(\frac{x}{r}\right) \right]_0^r$$

At $x = r$: the first term vanishes ($\sqrt{r^2 - r^2} = 0$), and $\sin^{-1}(1) = \pi/2$. At $x = 0$: both terms vanish.

$$A = \left(0 + \frac{r^2}{2} \cdot \frac{\pi}{2} \right) - 0 = \frac{\pi r^2}{4}$$

This is exactly one quadrant of the circle $x^2 + y^2 = r^2$. Since a circle has 4 congruent quadrants by symmetry, the full circle's area is

$$A_{\text{circle}} = 4 \times \frac{\pi r^2}{4} = \pi r^2 \quad \blacksquare$$

Final Answer

Quadrant area = $\frac{\pi r^2}{4}$ sq. units, hence full circle = πr^2

Common Mistake: Forgetting the standard integral formula for $\sqrt{r^2 - x^2}$ and trying a substitution from scratch under exam pressure – this formula is worth memorising outright.

Alternate Example – direct semicircle area

Source: Exercise 8, Q15

Given: $y = \sqrt{4 - x^2}$, x-axis (region: $-2 \leq x \leq 2$)

Solution:

This is the full upper semicircle of radius 2 (both quadrants), so integrate from -2 to 2 directly, or double the quadrant result with $r = 2$.

$$A = 2 \times \frac{\pi(2)^2}{4} = 2\pi$$

Final Answer

$A = 2\pi$ sq. units

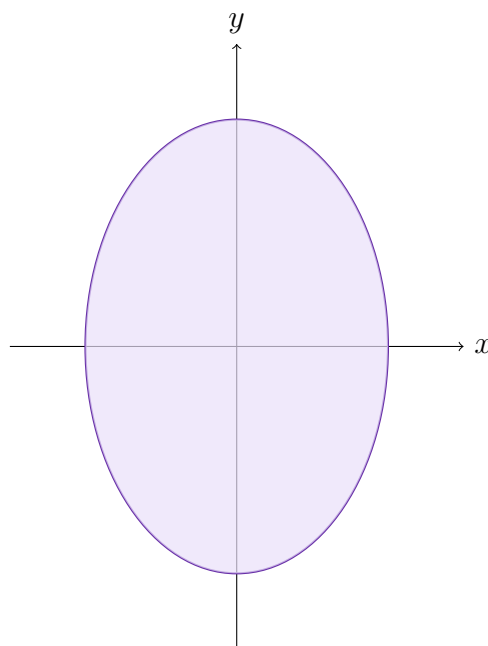
Quick Recall: $\int_0^r \sqrt{r^2 - x^2} dx = \frac{\pi r^2}{4}$ is a quarter-circle – memorise it, then just multiply by 2 or 4 depending on how much of the circle the question actually wants.

Type 13: Area under / enclosed by an ellipse

Standard ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Solve for y , integrate using the same $\sqrt{a^2 - x^2}$ -style formula (scaled by b/a), and use symmetry to go from a quadrant to the full ellipse.

Formula Used: $y = \frac{b}{a}\sqrt{a^2 - x^2}$, $A_{\text{full ellipse}} = \pi ab$

Source: Exercise 8, Q16



Ellipse $x^2/4 + y^2/9 = 1$

Given: $\frac{x^2}{4} + \frac{y^2}{9} = 1$

Solution:

Solve for y (upper half): $y = 3\sqrt{1 - \frac{x^2}{4}} = \frac{3}{2}\sqrt{4 - x^2}$. Part (i) asks for the area under this curve and above the x -axis, i.e. the full upper half of the ellipse (from $x = -2$ to $x = 2$).

$$A_{(i)} = \int_{-2}^2 \frac{3}{2}\sqrt{4 - x^2} dx = 2 \times \frac{3}{2} \int_0^2 \sqrt{4 - x^2} dx = 3 \times \frac{\pi(2)^2}{4} = 3\pi$$

For part (ii), the full ellipse enclosed area is exactly double the upper-half area by symmetry about the x -axis.

$$A_{(ii)} = 2 \times 3\pi = 6\pi$$

Final Answer

(i) $A = 3\pi$ sq. units (ii) $A = 6\pi$ sq. units

Common Mistake: Confusing which semi-axis is a and which is b – here $b = 3 > a = 2$, so the ellipse is taller than it is wide; always match the number under x^2 to a^2 and under y^2 to b^2 .

Alternate Board-Style Example – general form

Source: Exercise 8, Q17

Given: $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the general form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution:

Here $a = 4$, $b = 3$, so by the standard result $A = \pi ab$:

$$A_{(i)} = \pi(4)(3) = 12\pi$$

For the general ellipse, exactly the same integration (with a, b left as letters instead of numbers) gives the standard formula directly.

$$A_{(ii)} = \pi ab$$

Final Answer

$$(i) A = 12\pi \text{ sq. units} \quad (ii) A = \pi ab \text{ sq. units}$$

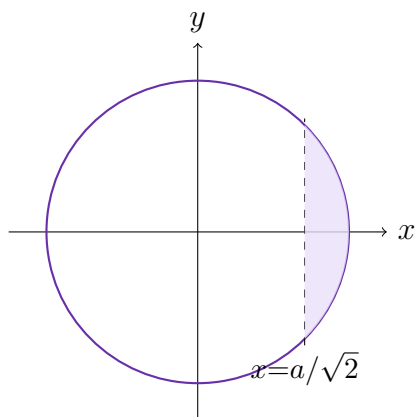
Quick Recall: An ellipse is a "stretched circle" – reuse the $\sqrt{a^2 - x^2}$ integral formula, scaled by b/a , and remember the full-ellipse shortcut $A = \pi ab$.

Type 14: Area of the minor segment of a circle

A chord (vertical line $x = c$) cuts a circle into two unequal pieces; the smaller piece (not containing the centre) is the minor segment. Integrate the circle's equation from the chord to the edge of the circle, doubling for both branches (above and below the x-axis).

Formula Used: $A_{\text{segment}} = 2 \int_c^a \sqrt{a^2 - x^2} dx$

Source: Exercise 8, Q18



Minor segment of $x^2 + y^2 = a^2$ cut by $x = a/\sqrt{2}$

Given: $x^2 + y^2 = a^2$, $x = \frac{a}{\sqrt{2}}$

Solution:

The minor segment is the sliver of the circle beyond the chord $x = a/\sqrt{2}$, symmetric

about the x-axis, so integrate the upper branch from $a/\sqrt{2}$ to a and double.

$$A = 2 \int_{a/\sqrt{2}}^a \sqrt{a^2 - x^2} dx = 2 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]_{a/\sqrt{2}}^a$$

At $x = a$: first term is 0; $\sin^{-1}(1) = \pi/2$, giving $\frac{a^2\pi}{4}$. At $x = a/\sqrt{2}$: $\sqrt{a^2 - a^2/2} = a/\sqrt{2}$, so the first term is $\frac{a}{2\sqrt{2}} \cdot \frac{a}{\sqrt{2}} = \frac{a^2}{4}$; and $\sin^{-1}(1/\sqrt{2}) = \pi/4$, giving $\frac{a^2\pi}{8}$ for the second term – total $\frac{a^2}{4} + \frac{a^2\pi}{8}$.

$$A = 2 \left[\frac{a^2\pi}{4} - \left(\frac{a^2}{4} + \frac{a^2\pi}{8} \right) \right] = 2 \left[\frac{a^2\pi}{8} - \frac{a^2}{4} \right] = \frac{a^2\pi}{4} - \frac{a^2}{2} = \frac{a^2}{4}(\pi - 2)$$

Final Answer

$$A = \frac{a^2}{4}(\pi - 2) \text{ sq. units}$$

Common Mistake: Computing the *major* segment (the larger piece, containing the centre) instead of the minor one – always sketch first to see which side of the chord is being asked for.

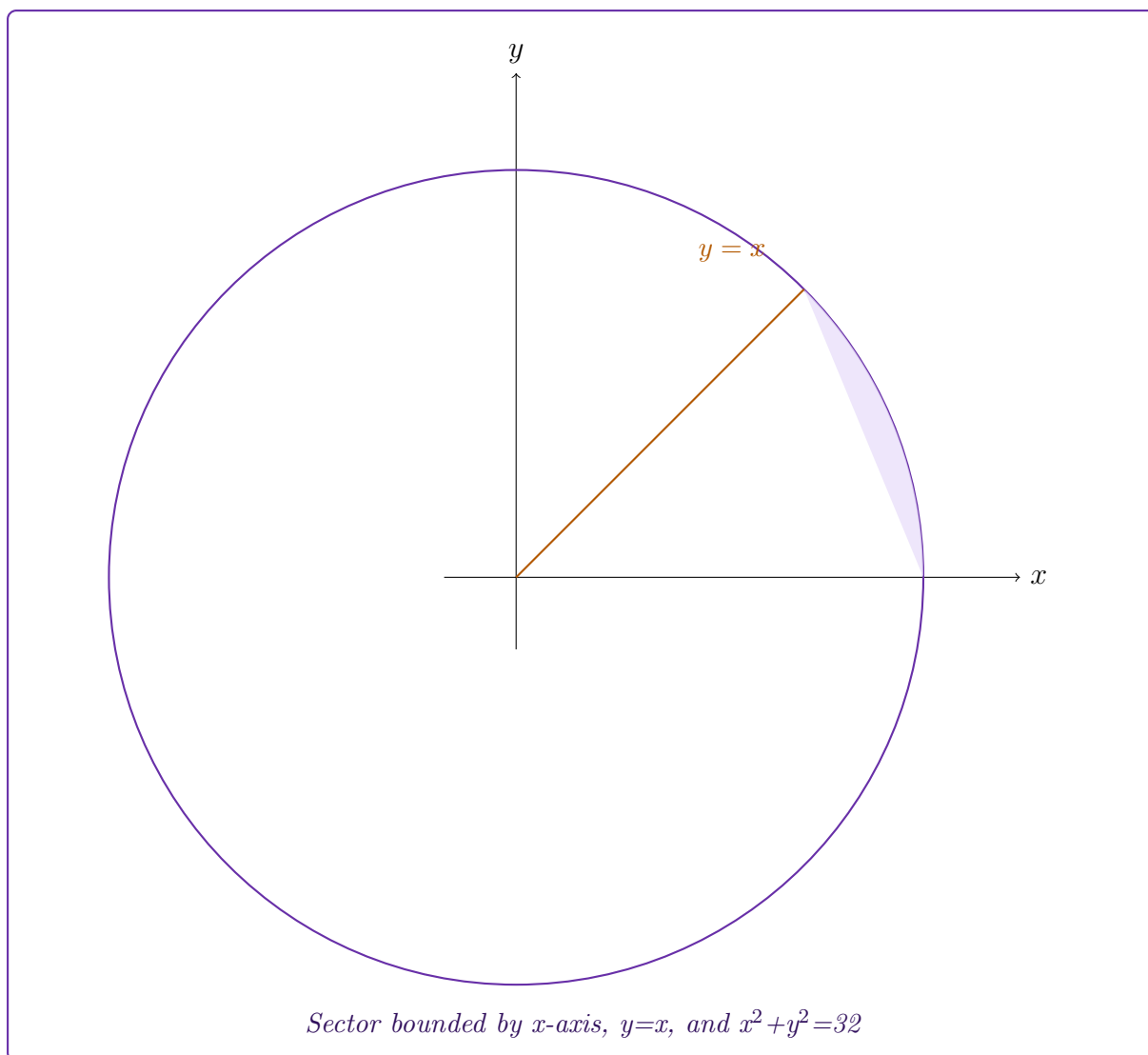
Quick Recall: Minor segment = integrate from the chord out to the circle's edge, using the standard $\sqrt{a^2 - x^2}$ formula, doubled for symmetry about the x-axis.

Type 15: Sector-type region – circle, a line through the origin, and an axis

The region in the first quadrant is bounded by an axis, a line $y = mx$ through the origin, and a circular arc – geometrically this is a circular sector. Split the integral at the point where the line meets the circle: integrate under the line up to that point, then under the circle from there to the axis. (Shortcut check: this always equals the sector formula $\frac{1}{2}r^2\theta$, where θ is the angle the line makes with the axis – useful for verifying your integration answer.)

Formula Used: $A = \int_0^{x_0} mx dx + \int_{x_0}^r \sqrt{r^2 - x^2} dx$ ($= \frac{1}{2}r^2\theta$ as a check)

Source: Exercise 8, Q19(i)



Given: $x^2 + y^2 = 32$, $y = x$, x-axis, first quadrant

Solution:

Find where the line meets the circle: substitute $y = x$ into $x^2 + y^2 = 32 \Rightarrow 2x^2 = 32 \Rightarrow x = 4$ (taking the positive root), so the point is $(4, 4)$.

$$A = \underbrace{\int_0^4 x \, dx}_{\text{under the line}} + \underbrace{\int_4^{4\sqrt{2}} \sqrt{32 - x^2} \, dx}_{\text{under the circle}}$$

The first piece: $\left[\frac{x^2}{2} \right]_0^4 = 8$. For the second piece, use the standard circle-integral formula with $r^2 = 32$:

$$\int_4^{4\sqrt{2}} \sqrt{32 - x^2} \, dx = \left[\frac{x}{2} \sqrt{32 - x^2} + 16 \sin^{-1} \left(\frac{x}{4\sqrt{2}} \right) \right]_4^{4\sqrt{2}}$$

At $x = 4\sqrt{2}$: first term 0; $\sin^{-1}(1) = \pi/2$, giving 8π . At $x = 4$: $\sqrt{32 - 16} = 4$, so first

term is $\frac{4}{2} \times 4 = 8$; $\sin^{-1}(1/\sqrt{2}) = \pi/4$, giving $16 \times \pi/4 = 4\pi$ – total $8 + 4\pi$.

$$\int_4^{4\sqrt{2}} \sqrt{32 - x^2} dx = 8\pi - (8 + 4\pi) = 4\pi - 8$$

$$A = 8 + (4\pi - 8) = 4\pi$$

Final Answer

$A = 4\pi$ sq. units (check: sector angle $45^\circ = \pi/4$, $\frac{1}{2}(32)(\pi/4) = 4\pi$ ✓)

Common Mistake: Integrating the circle over the wrong range, or forgetting to split the integral at the line-circle intersection point at all.

Note – how this differs: Q19(ii)-(iv) and Q20 below are the same sector idea at different angles – (ii) and (iii) give the sector directly via the shortcut once the angle is found, and Q20 runs the type in reverse: the area is given and you solve for the slope m .

Sub-part: sector via the angle shortcut

Source: Exercise 8, Q19(iv)

Given: $y = \sqrt{3}x$, $y = \sqrt{4 - x^2}$, x-axis, first quadrant

Solution:

The line $y = \sqrt{3}x$ makes angle θ with the x-axis where $\tan \theta = \sqrt{3}$, so $\theta = 60^\circ = \pi/3$. The circle has radius $r = 2$. Using the sector formula (which the same split-integral method above always reduces to):

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(4) \left(\frac{\pi}{3}\right) = \frac{2\pi}{3}$$

Final Answer

$$A = \frac{2\pi}{3} \text{ sq. units}$$

Sub-part: reverse problem – find the slope

Source: Exercise 8, Q20

Given: $y = mx$ ($m > 0$), $x^2 + y^2 = 4$, x-axis, first quadrant, $A = \frac{\pi}{2}$

Solution:

Let θ be the angle the line makes with the x-axis, so $m = \tan \theta$. By the sector formula with $r^2 = 4$:

$$\begin{aligned} \frac{1}{2}(4)\theta &= \frac{\pi}{2} \Rightarrow 2\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4} \\ m &= \tan\left(\frac{\pi}{4}\right) = 1 \end{aligned}$$

Final Answer

$$m = 1$$

Common Mistake: Trying to solve for m by re-doing the full split integral with m left as a letter – much faster to reduce to the sector-angle formula once you recognise the shape.

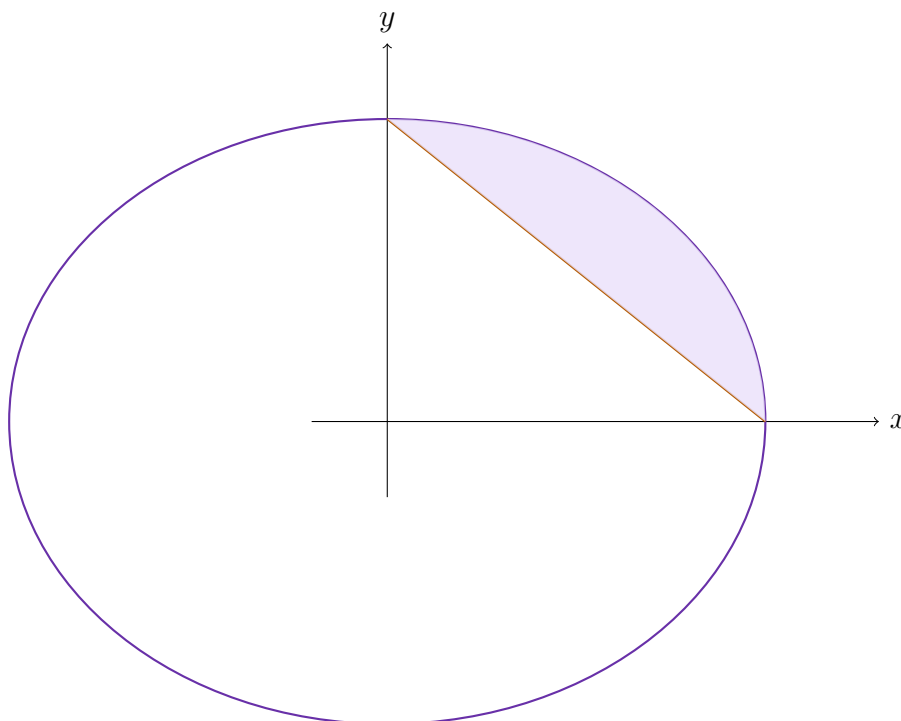
Quick Recall: A region bounded by an axis, a line through the origin, and a circle is a sector – integrate line-then-circle if asked to "using integration", but the sector formula $\frac{1}{2}r^2\theta$ is the fast check (or fast solve, if the angle/slope is the unknown).

Type 16: Smaller region common to an ellipse and a line

The line joins the two axis-intercepts of the ellipse (i.e. it is the chord connecting the ends of the semi-major and semi-minor axes). The smaller enclosed region, between the chord and the nearer elliptical arc, equals (quarter-ellipse area) – (triangle area) – no integration of the chord itself is needed once you see this.

Formula Used: $A = \frac{\pi ab}{4} - \frac{1}{2}ab$

Source: Exercise 8, Q21



Ellipse $x^2/25 + y^2/16 = 1$ and chord $x/5 + y/4 = 1$

Given: $\frac{x^2}{25} + \frac{y^2}{16} = 1, \quad \frac{x}{5} + \frac{y}{4} = 1$

Solution:

The ellipse has $a = 5$, $b = 4$. The line's intercepts are $(5, 0)$ and $(0, 4)$ – exactly the ends of the ellipse's axes – so this line is the chord joining them, and the region "smaller" than the chord (between the chord and the arc) is the quarter-ellipse minus the right-triangle it cuts off.

$$A_{\text{quarter ellipse}} = \frac{\pi(5)(4)}{4} = 5\pi, \quad A_{\triangle} = \frac{1}{2}(5)(4) = 10$$

$$A = 5\pi - 10 = 5(\pi - 2)$$

Final Answer

$$A = 5(\pi - 2) \text{ sq. units}$$

Common Mistake: Trying to integrate the ellipse-minus-line difference directly with the messy $\sqrt{a^2 - x^2}$ formula – recognising the chord joins the axis-endpoints (so the region is a clean "sector minus triangle") saves a lot of algebra.

Quick Recall: If a line joins the two intercepts of an ellipse, the smaller region cut off is just quarter-ellipse minus the right triangle – no messy integral needed.

Type 17: Segment of an ellipse (bounded by an arc and its chord)

Essentially the same computation as Type 16, but phrased as "arc AB and chord AB" with points A and B given via their distances from the centre along each axis (OA, OB) rather than via a line equation. Identify $a = OA$, $b = OB$, and apply the same quarter-ellipse-minus-triangle idea.

Formula Used: $A = \frac{\pi ab}{4} - \frac{1}{2}ab$

Source: Exercise 8, Q22

Given: $9x^2 + y^2 = 36$ (i.e. $\frac{x^2}{4} + \frac{y^2}{36} = 1$), $OA = 2$, $OB = 6$

Solution:

Rewriting the ellipse in standard form gives $a = 2$ (x-intercept, matching $OA = 2$) and $b = 6$ (y-intercept, matching $OB = 6$) – so A and B are exactly the ellipse's axis endpoints, and AB is the chord joining them, just as in Type 16.

$$A_{\text{quarter ellipse}} = \frac{\pi(2)(6)}{4} = 3\pi, \quad A_{\triangle OAB} = \frac{1}{2}(2)(6) = 6$$

$$A = 3\pi - 6 = 3(\pi - 2)$$

Final Answer

$$A = 3(\pi - 2) \text{ sq. units}$$

Common Mistake: Not converting $9x^2 + y^2 = 36$ to standard form first – dividing through by 36 to get $\frac{x^2}{4} + \frac{y^2}{36} = 1$ is essential before reading off a and b .

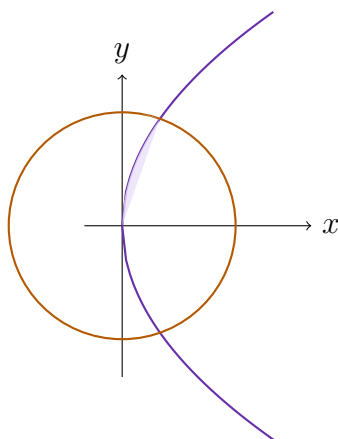
Quick Recall: "Arc + chord" is the same picture as "ellipse + line joining intercepts" from Type 16 – always reduce to quarter-ellipse minus triangle.

Type 18: Region common to a circle and another curve

The region satisfies two inequalities simultaneously – one from a circle, one from a parabola (or a second circle-type inequality). Find the intersection point(s) of the two boundary curves first, then split the integral: use whichever curve gives the *smaller* height on each sub-interval (since the region must lie inside both).

Formula Used: $A = 2 \left[\int_0^{x_0} y_{\text{curve 1}} dx + \int_{x_0}^{x_1} y_{\text{curve 2}} dx \right]$

Source: Exercise 8, Q27(i)



Region common to $y^2 \leq 4x$ and $4x^2 + 4y^2 \leq 9$

Given: $y^2 \leq 4x$ (parabola), $4x^2 + 4y^2 \leq 9$ (circle $x^2 + y^2 \leq \frac{9}{4}$, $r = \frac{3}{2}$)

Solution:

Find where $y^2 = 4x$ meets $x^2 + y^2 = \frac{9}{4}$: substitute $y^2 = 4x$ into the circle equation.

$$x^2 + 4x = \frac{9}{4} \Rightarrow 4x^2 + 16x - 9 = 0 \Rightarrow x = \frac{-16 \pm \sqrt{256 + 144}}{8} = \frac{-16 \pm 20}{8}$$

Taking the positive root, $x = \frac{1}{2}$ (giving $y = \pm\sqrt{2}$). For $0 \leq x \leq \frac{1}{2}$ the parabola is the tighter (smaller) bound; for $\frac{1}{2} \leq x \leq \frac{3}{2}$ the circle is tighter.

Case: $0 \leq x \leq \frac{1}{2}$ (bounded by the parabola)

$$\int_0^{1/2} 2\sqrt{x} \, dx = \left[\frac{4}{3} x^{3/2} \right]_0^{1/2} = \frac{4}{3} \cdot \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{3}$$

Case: $\frac{1}{2} \leq x \leq \frac{3}{2}$ (bounded by the circle)

$$\int_{1/2}^{3/2} \sqrt{\frac{9}{4} - x^2} \, dx = \left[\frac{x}{2} \sqrt{\frac{9}{4} - x^2} + \frac{9}{8} \sin^{-1} \left(\frac{2x}{3} \right) \right]_{1/2}^{3/2}$$

At $x = \frac{3}{2}$: first term 0, $\sin^{-1}(1) = \pi/2$, giving $\frac{9\pi}{16}$. At $x = \frac{1}{2}$: $\sqrt{\frac{9}{4} - \frac{1}{4}} = \sqrt{2}$, first term $= \frac{1}{4}\sqrt{2}$; $\sin^{-1}(\frac{1}{3})$ gives $\frac{9}{8} \sin^{-1}(\frac{1}{3})$.

$$= \frac{9\pi}{16} - \frac{\sqrt{2}}{4} - \frac{9}{8} \sin^{-1} \left(\frac{1}{3} \right)$$

Adding both pieces (this is the upper-half area) and doubling for the lower half:

$$A = 2 \left[\frac{\sqrt{2}}{3} + \frac{9\pi}{16} - \frac{\sqrt{2}}{4} - \frac{9}{8} \sin^{-1} \left(\frac{1}{3} \right) \right] = \frac{\sqrt{2}}{6} + \frac{9\pi}{8} - \frac{9}{4} \sin^{-1} \left(\frac{1}{3} \right)$$

Final Answer

$$A = \frac{\sqrt{2}}{6} + \frac{9\pi}{8} - \frac{9}{4} \sin^{-1} \left(\frac{1}{3} \right) \text{ sq. units}$$

Common Mistake: Using the larger of the two curve-heights instead of the smaller one – the region common to two inequalities is always bounded by whichever curve is tighter at that x-value.

Alternate Example – circle and parabola, cleaner numbers

Source: Exercise 8, Q28

$$\text{Given: } y^2 = 2x, \quad x^2 + y^2 = 4x \text{ (i.e. } (x-2)^2 + y^2 = 4)$$

Solution:

Substituting $y^2 = 2x$ into the circle's equation: $x^2 + 2x = 4x \Rightarrow x^2 - 2x = 0 \Rightarrow x = 0, 2$. For $0 \leq x \leq 2$, test $x = 1$: parabola gives $y = \sqrt{2} \approx 1.41$, circle gives $y = \sqrt{4-1} = \sqrt{3} \approx 1.73$ – so the parabola is the tighter (lower) curve throughout, and the smaller common region is simply the strip between them.

$$A = 2 \int_0^2 \left[\sqrt{4x - x^2} - \sqrt{2x} \right] dx$$

$$\int_0^2 \sqrt{2x} \, dx = \sqrt{2} \cdot \frac{2}{3} x^{3/2} \Big|_0^2 = \frac{2}{3} \times 4 = \frac{8}{3}. \text{ For the circle piece, substitute } u = x - 2:$$

$$\int_0^2 \sqrt{4 - (x-2)^2} \, dx = \int_{-2}^0 \sqrt{4 - u^2} \, du$$

$$= \left[\frac{u}{2} \sqrt{4 - u^2} + 2 \sin^{-1} \left(\frac{u}{2} \right) \right]_{-2}^0 = 0 - (0 + 2 \sin^{-1}(-1)) = -2 \left(-\frac{\pi}{2} \right) = \pi$$

$$A = 2 \left[\pi - \frac{8}{3} \right] = 2\pi - \frac{16}{3} = 2 \left(\pi - \frac{8}{3} \right)$$

Final Answer

$$A = 2 \left(\pi - \frac{8}{3} \right) \text{ sq. units}$$

Quick Recall: "Region common to two curves" means: find where they cross, then on each piece use whichever curve gives the smaller height – sketch it first so you don't need to guess.

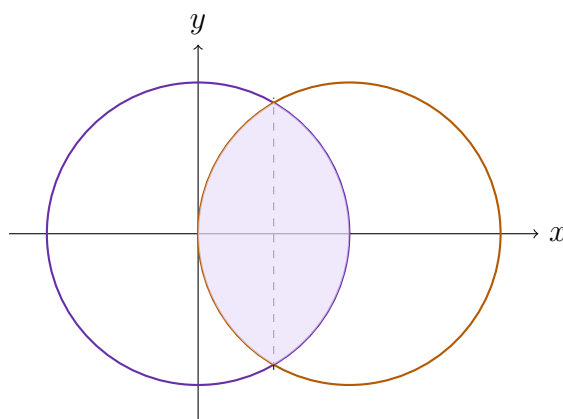
Type 19: Region common to two intersecting circles

Two circles of equal or different radii overlap in a lens-shaped region symmetric about the perpendicular bisector of the line joining their centres. Find the intersection points (subtract the two circle equations – this eliminates the squared terms and gives a straight vertical line of symmetry directly), then integrate each circle's arc on its own side of that line and add.

Formula Used: $A = 2 \left[\int_{x_1}^{x_m} \sqrt{r_1^2 - x^2} dx + \int_{x_m}^{x_2} \sqrt{r_2^2 - (x - d)^2} dx \right]$

Source: Exercise 8, Q26(i)

★ **Board Favorite** – Two-circles-overlap questions are a recurring ISC favourite because subtracting the two circle equations to find the axis of symmetry is a clean, testable trick that shows up almost every year in some form.



Circles $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$

Given: $x^2 + y^2 = 4$, $(x - 2)^2 + y^2 = 4$

Solution:

Subtracting the two equations eliminates y^2 and the x^2 terms:

$$x^2 - (x - 2)^2 = 0 \Rightarrow 4x - 4 = 0 \Rightarrow x = 1$$

This vertical line $x = 1$ is the axis of symmetry of the overlap region. By symmetry, the left half (from the first circle) equals the right half (from the second circle), and the whole figure is also symmetric top-to-bottom about the x-axis.

$$A = 2 \left[\int_0^1 \sqrt{4 - (x - 2)^2} dx + \int_1^2 \sqrt{4 - x^2} dx \right]$$

For the second piece (standard form directly):

$$\int_1^2 \sqrt{4 - x^2} dx = \left[\frac{x}{2} \sqrt{4 - x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) \right]_1^2 = \pi - \left(\frac{\sqrt{3}}{2} + \frac{\pi}{3} \right) = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

For the first piece, substitute $u = x - 2$ (so the limits become $u = -2$ to $u = -1$):

$$\int_{-2}^{-1} \sqrt{4 - u^2} du = \left[\frac{u}{2} \sqrt{4 - u^2} + 2 \sin^{-1} \left(\frac{u}{2} \right) \right]_{-2}^{-1} = \left(-\frac{\sqrt{3}}{2} - \frac{\pi}{3} \right) - (-\pi) = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

Both pieces come out identical, as symmetry predicts.

$$A = 2 \left[\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) + \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \right] = 2 \left[\frac{4\pi}{3} - \sqrt{3} \right] = \frac{8\pi}{3} - 2\sqrt{3}$$

Final Answer

$$A = \frac{8\pi}{3} - 2\sqrt{3} \text{ sq. units}$$

Common Mistake: Trying to integrate one circle's equation across the *entire* overlap region – each circle's own equation is only valid as the true boundary on its own side of the symmetry line; using it beyond that overshoots the actual region.

Alternate Board-Style Example

Source: Exercise 8, Q26(ii)

$$\text{Given: } x^2 + y^2 = 1, \quad (x - 1)^2 + y^2 = 1$$

Solution:

Subtracting: $x^2 - (x - 1)^2 = 0 \Rightarrow 2x - 1 = 0 \Rightarrow x = \frac{1}{2}$, the symmetry line. By the identical method (radius 1, centre separation 1):

$$A = 2 \left[\int_0^{1/2} \sqrt{1 - (x - 1)^2} dx + \int_{1/2}^1 \sqrt{1 - x^2} dx \right]$$

$$\int_{1/2}^1 \sqrt{1 - x^2} dx = \left[\frac{x}{2} \sqrt{1 - x^2} + \frac{1}{2} \sin^{-1} x \right]_{1/2}^1 = \frac{\pi}{4} - \left(\frac{\sqrt{3}}{8} + \frac{\pi}{12} \right) = \frac{\pi}{6} - \frac{\sqrt{3}}{8}$$

By the same symmetry as before, the other piece is identical, so

$$A = 2 \left[2 \left(\frac{\pi}{6} - \frac{\sqrt{3}}{8} \right) \right] = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

Final Answer

$$A = \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \text{ sq. units}$$

Quick Recall: Subtract the two circle equations first – the x^2 (and y^2) terms cancel and what's left is the vertical line of symmetry; integrate each circle on its own side and use symmetry to halve your work.

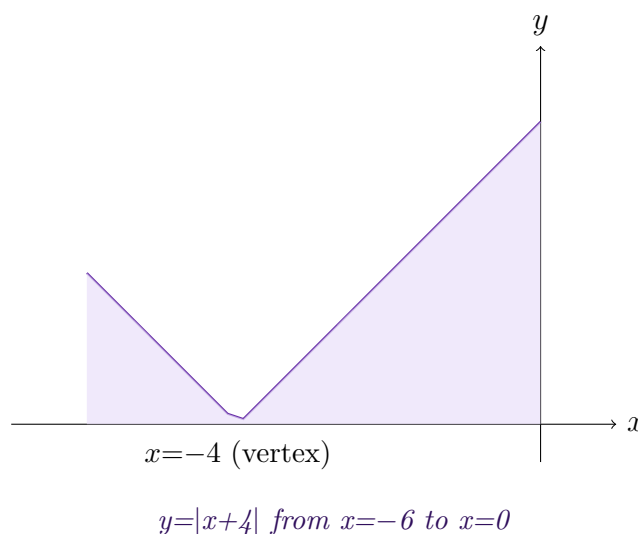
IV. Modulus, Exponential/Log, and Direct Inequality Regions

Type 20: Regions bounded by modulus-function curves

The boundary involves $|x - a|$ directly. Split the domain at the point where the expression inside the modulus changes sign (i.e. where it equals zero), rewrite the modulus as a plain linear expression on each side, and integrate each piece separately – never integrate straight through a modulus sign without splitting first.

Formula Used: $|x - a| = \begin{cases} -(x - a), & x < a \\ (x - a), & x \geq a \end{cases}$

Source: Exercise 8, Q31(i)



Given: $y = |x + 4|$, $x = -6$, $x = 0$, x-axis

Solution:

The expression $x + 4$ changes sign at $x = -4$, which lies inside $[-6, 0]$, so split there.

Case: $-6 \leq x \leq -4$ (here $x + 4 < 0$, so $|x + 4| = -(x + 4)$)

$$\int_{-6}^{-4} -(x + 4) dx = \left[-\frac{x^2}{2} - 4x \right]_{-6}^{-4} = (-8 + 16) - (-18 + 24) = 8 - 6 = 2$$

Case: $-4 \leq x \leq 0$ (here $x + 4 \geq 0$, so $|x + 4| = x + 4$)

$$\int_{-4}^0 (x+4) dx = \left[\frac{x^2}{2} + 4x \right]_{-4}^0 = 0 - (8 - 16) = 8$$

$$A = 2 + 8 = 10$$

Final Answer

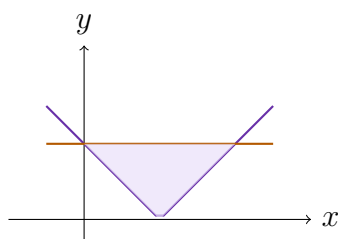
$A = 10$ sq. units

Common Mistake: Integrating $|x+4|$ as if it were simply $x+4$ across the whole interval without splitting – this silently gives a wrong (usually smaller, sometimes even negative-looking) answer whenever the interval straddles the vertex.

Note – how this differs: Sub-part (b) is a region trapped *between* two curves, one of which is a modulus function and the other a horizontal line – same splitting idea, but now both boundaries need comparing on each side of the vertex.

Sub-part: modulus curve vs. a horizontal line

Source: Exercise 8, Q31(ii)



Region between $y=|x-1|$ and $y=1$

Given: $y = |x-1|$, $y = 1$

Solution:

Find where they meet: $|x-1| = 1 \Rightarrow x-1 = \pm 1 \Rightarrow x = 0, 2$. On $(0, 2)$, test $x = 1$: the curve gives 0, the line gives 1 – the line is on top. By symmetry about $x = 1$, compute one half and double.

$$A = 2 \int_0^1 [1 - (1-x)] dx = 2 \int_0^1 x dx = 2 \left[\frac{x^2}{2} \right]_0^1 = 1$$

Final Answer

$A = 1$ sq. unit

Note – how this differs: Sub-part (c) replaces the horizontal line with a second curve, $y = (x-1)^2$ – same symmetry trick about the vertex, but now both boundaries are curved.

Sub-part: two curves through the same vertex

Source: Exercise 8, Q32

Given: $y = (x - 1)^2$, $y = |x - 1|$

Solution:

Let $t = x - 1$ to see the symmetry clearly: the curves are $y = t^2$ and $y = |t|$, both passing through the origin (in t). They meet where $t^2 = |t| \Rightarrow |t|(|t| - 1) = 0 \Rightarrow t = 0, \pm 1$, i.e. $x = 0, 1, 2$. Testing $t = 0.5$: $|t| = 0.5 > t^2 = 0.25$, so $|t|$ is on top on both sides by symmetry.

$$A = 2 \int_0^1 (t - t^2) dt = 2 \left[\frac{t^2}{2} - \frac{t^3}{3} \right]_0^1 = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = 2 \times \frac{1}{6} = \frac{1}{3}$$

Final Answer

$$A = \frac{1}{3} \text{ sq. units}$$

Common Mistake: Forgetting that shifting the variable ($t = x - 1$) doesn't change the region's shape or area – it only makes the symmetry easier to see; the final numeric answer is unaffected.

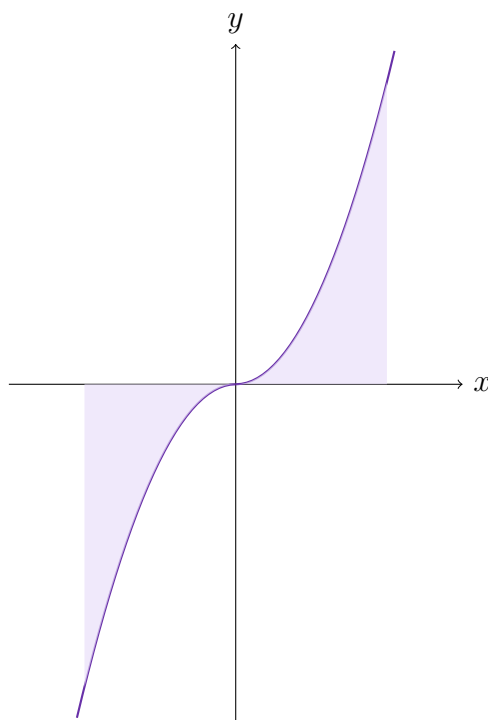
Quick Recall: Modulus curves always need splitting at the point where the inside expression is zero – look for a shift-symmetry around that point to cut your work in half.

Type 21: Area bounded by $y = x|x|$

This function is a "signed parabola" – it equals x^2 for $x \geq 0$ and $-x^2$ for $x < 0$, so it is odd and passes smoothly through the origin (no corner, unlike a plain modulus function). Split at $x = 0$ and remember that on the negative side the curve dips below the axis, so that piece's integral must be taken as a positive area.

Formula Used: $y = x|x| = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$

Source: Exercise 8, Q37



$y = x|x|$ from $x = -2$ to $x = 2$

Given: $y = x|x|$, $x = -2$, $x = 2$

Solution:

Split at $x = 0$ where the definition of y changes.

Case: $0 \leq x \leq 2$ (here $y = x^2 \geq 0$)

$$\int_0^2 x^2 dx = \left[\frac{x^3}{3} \right]_0^2 = \frac{8}{3}$$

Case: $-2 \leq x \leq 0$ (here $y = -x^2 \leq 0$ – curve is below the axis)

$$\int_{-2}^0 (-x^2) dx = \left[-\frac{x^3}{3} \right]_{-2}^0 = 0 - \frac{8}{3} = -\frac{8}{3}$$

This integral is negative because the curve lies below the x-axis; the actual area contributed is its absolute value, $\frac{8}{3}$.

$$A = \frac{8}{3} + \frac{8}{3} = \frac{16}{3}$$

Final Answer

$$A = \frac{16}{3} \text{ sq. units}$$

Common Mistake: Integrating $x|x|$ across -2 to 2 in one shot without splitting the definition – because the function is odd, a careless single integral over a symmetric interval gives zero (the positive and negative parts cancel), which is not the area.

Quick Recall: $x|x|$ is x^2 on the right and $-x^2$ on the left – split at zero, and remember the left piece needs its sign flipped to become a positive area.

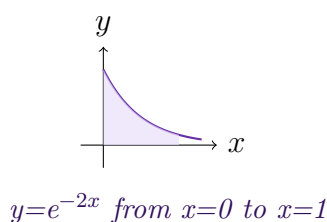
Type 22: Area under exponential or logarithmic curves

The boundary is e^{kx} , $\ln x$, or a difference of two such curves. Use $\int e^{kx} dx = \frac{1}{k}e^{kx}$ directly; for $\ln x$, integrate by parts with the standard result $\int \ln x dx = x \ln x - x$. For a region between two exponentials, always test a point first to see which curve is on top (exponentials cross less predictably than polynomials).

Formula Used: $\int e^{kx} dx = \frac{1}{k}e^{kx} + C$

Formula Used: $\int \ln x dx = x \ln x - x + C$

Source: Exercise 8, Q35(i)



Given: $y = e^{-2x}$, $x = 0$, $x = 1$

Solution:

$$A = \int_0^1 e^{-2x} dx = \left[-\frac{1}{2}e^{-2x} \right]_0^1 = -\frac{1}{2}e^{-2} - \left(-\frac{1}{2} \right) = \frac{1}{2}(1 - e^{-2})$$

Final Answer

$$A = \frac{1}{2}(1 - e^{-2}) \text{ sq. units}$$

Common Mistake: Sign errors differentiating/integrating e^{-2x} – remember the chain-rule factor $\frac{1}{k}$ picks up a *negative* sign when k itself is negative.

Sub-part: logarithmic curve

Source: Exercise 8, Q35(ii)

Given: $y = \ln x$, $x = 1$, $x = 2$

Solution:

$$A = \int_1^2 \ln x \, dx = [x \ln x - x]_1^2 = (2 \ln 2 - 2) - (1 \cdot 0 - 1) = 2 \ln 2 - 2 + 1$$

$$A = 2 \ln 2 - 1$$

Final Answer

$$A = 2 \ln 2 - 1 \text{ sq. units}$$

Sub-part: area between two exponentials

Source: Exercise 8, Q35(iii)

Given: $y = e^x$, $y = e^{3x}$, $x = 0$, $x = 1$

Solution:

Test $x = 0.5$: $e^{3(0.5)} = e^{1.5} \approx 4.48$ is larger than $e^{0.5} \approx 1.65$, so e^{3x} is on top for $x > 0$.

$$A = \int_0^1 (e^{3x} - e^x) \, dx = \left[\frac{e^{3x}}{3} - e^x \right]_0^1 = \left(\frac{e^3}{3} - e \right) - \left(\frac{1}{3} - 1 \right)$$

$$A = \frac{e^3}{3} - e - \frac{1}{3} + 1 = \frac{e^3}{3} - e + \frac{2}{3} = \frac{1}{3} (e^3 - 3e + 2)$$

Final Answer

$$A = \frac{1}{3} (e^3 - 3e + 2) \text{ sq. units}$$

Sub-part: log curve against the x-axis

Source: Exercise 8, Q35(iv)

Given: $y = \ln x$, x-axis, $x = 1$, $x = e^2$

Solution:

$$A = \int_1^{e^2} \ln x \, dx = [x \ln x - x]_1^{e^2} = (e^2 \cdot 2 - e^2) - (0 - 1) = e^2 + 1$$

Final Answer

$$A = e^2 + 1 \text{ sq. units}$$

Sub-part: exponential vs. reciprocal curve

Source: Exercise 8, Q35(v)

Given: $y = e^{-x}$, $y = \frac{1}{x}$, $x = 1$, $x = 2$

Solution:

Test $x = 1$: $\frac{1}{x} = 1$ while $e^{-x} \approx 0.37$, so $\frac{1}{x}$ is the upper curve throughout $[1, 2]$ (it stays larger than the fast-decaying exponential here).

$$A = \int_1^2 \left(\frac{1}{x} - e^{-x} \right) dx = [\ln x + e^{-x}]_1^2 = (\ln 2 + e^{-2}) - (0 + e^{-1})$$

$$A = \ln 2 + e^{-2} - e^{-1}$$

Final Answer

$$A = \ln 2 + e^{-2} - e^{-1} \text{ sq. units}$$

Common Mistake: Assuming e^{-x} is always the smaller function just because it "decays" – always check with a test point; here $1/x$ stays on top throughout the given interval.

Quick Recall: Exponentials integrate with a $1/k$ factor, logs integrate by parts to $x \ln x - x$ – and always test a point before assuming which exponential curve sits on top.

Type 23: Region defined directly by set-builder inequalities

No curves are named in words – the region is handed to you as a set $\{(x, y) : \dots\}$ with two or more inequalities. Translate each inequality into "above/below this curve" first, sketch mentally (or on paper), find where the bounding curves cross, and integrate whichever bound is tighter (smaller) on each piece.

Formula Used: $A = \int [\min(\text{upper bounds}) - \max(\text{lower bounds})]$

Source: Exercise 8, Q33(i)

Given: $(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2$

Solution:

The upper bound on y is whichever of $x^2 + 1$ and $x + 1$ is smaller at each x . Find where they cross: $x^2 + 1 = x + 1 \Rightarrow x^2 = x \Rightarrow x = 0, 1$. Testing $x = 0.5$: $x^2 + 1 = 1.25 < x + 1 = 1.5$, so the parabola is the tighter bound on $[0, 1]$; testing $x = 1.5$: $x^2 + 1 = 3.25 > x + 1 = 2.5$, so the line is tighter on $[1, 2]$.

Case: $0 \leq x \leq 1$ (bound: $x^2 + 1$)

$$\int_0^1 (x^2 + 1) dx = \left[\frac{x^3}{3} + x \right]_0^1 = \frac{1}{3} + 1 = \frac{4}{3}$$

Case: $1 \leq x \leq 2$ (bound: $x + 1$)

$$\int_1^2 (x + 1) dx = \left[\frac{x^2}{2} + x \right]_1^2 = (2 + 2) - \left(\frac{1}{2} + 1 \right) = \frac{5}{2}$$

$$A = \frac{4}{3} + \frac{5}{2} = \frac{8}{6} + \frac{15}{6} = \frac{23}{6}$$

Final Answer

$$A = \frac{23}{6} \text{ sq. units}$$

Common Mistake: Adding both upper bounds together, or integrating the *larger* of the two instead of the smaller – the inequalities $y \leq x^2 + 1$ AND $y \leq x + 1$ both restrict y , so the true ceiling is whichever is lower.

Alternate Board-Style Example

Source: Exercise 8, Q33(ii)

Given: $(x, y) : 0 \leq 2y \leq x^2, 0 \leq y \leq x, 0 \leq x \leq 3$

Solution:

Rewrite the first inequality as $y \leq \frac{x^2}{2}$. Find where $\frac{x^2}{2} = x \Rightarrow x^2 = 2x \Rightarrow x = 0, 2$.

Testing $x = 1$: $\frac{x^2}{2} = 0.5 < x = 1$, so the parabola is tighter on $[0, 2]$; testing $x = 2.5$: $\frac{x^2}{2} = 3.125 > x = 2.5$, so the line is tighter on $[2, 3]$.

Case: $0 \leq x \leq 2$ (bound: $x^2/2$)

$$\int_0^2 \frac{x^2}{2} dx = \left[\frac{x^3}{6} \right]_0^2 = \frac{8}{6} = \frac{4}{3}$$

Case: $2 \leq x \leq 3$ (bound: x)

$$\int_2^3 x dx = \left[\frac{x^2}{2} \right]_2^3 = \frac{9}{2} - 2 = \frac{5}{2}$$

$$A = \frac{4}{3} + \frac{5}{2} = \frac{23}{6}$$

Final Answer

$$A = \frac{23}{6} \text{ sq. units}$$

A Related Flavour: inequality region formed by a circle and a line

Source: Exercise 8, Q30

Given: $(x, y) : x^2 + y^2 \leq 1, \quad x + y \geq 1$

Solution:

$x^2 + y^2 \leq 1$ is the inside of the unit circle; $x + y \geq 1$ is the far side of the chord $x + y = 1$ (away from the origin, since the origin gives $0 < 1$). So the region is the minor segment of the unit circle cut off by this chord – the same idea as Type 14, but now the chord is diagonal. Using the segment formula with $r = 1$ and perpendicular distance from centre $d = \frac{1}{\sqrt{2}}$ (distance from origin to the line $x + y = 1$):

$$\begin{aligned} A &= r^2 \cos^{-1}\left(\frac{d}{r}\right) - d\sqrt{r^2 - d^2} = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) - \frac{1}{\sqrt{2}}\sqrt{1 - \frac{1}{2}} \\ &= \frac{\pi}{4} - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{\pi}{4} - \frac{1}{2} = \frac{\pi - 2}{4} \end{aligned}$$

Final Answer

$$A = \frac{\pi - 2}{4} \text{ sq. units}$$

Quick Recall: Set-builder regions are just curves in disguise – rewrite every inequality as " $y \leq \dots$ " or " $y \geq \dots$ ", find the crossover point(s), and always use the tighter bound.

Type 24: Equal-area division – find the unknown

A total area is given (or easily computed), and a vertical ordinate $x = a$ is stated to split it into two equal halves; find a . Compute the total area first, set it equal to twice the area of one piece (expressed with a as the upper or lower limit), and solve the resulting equation for a .

Formula Used: $\int_{x_1}^a y \, dx = \frac{1}{2} \int_{x_1}^{x_2} y \, dx$

Source: Exercise 8, Q39

Given: $y = 1 + \frac{8}{x^2}$, x-axis, $x = 2$, $x = 4$, ordinate $x = a$ bisects the area

Solution:

First find the total area between $x = 2$ and $x = 4$.

$$A_{\text{total}} = \int_2^4 \left(1 + \frac{8}{x^2}\right) dx = \left[x - \frac{8}{x}\right]_2^4 = (4 - 2) - (2 - 4) = 2 - (-2) = 4$$

Half of this total is 2. Now find a such that the area from $x = 2$ to $x = a$ equals 2.

$$\int_2^a \left(1 + \frac{8}{x^2}\right) dx = \left[x - \frac{8}{x}\right]_2^a = \left(a - \frac{8}{a}\right) - (2 - 4) = a - \frac{8}{a} + 2$$

Set this equal to 2:

$$a - \frac{8}{a} + 2 = 2 \Rightarrow a - \frac{8}{a} = 0 \Rightarrow a^2 = 8 \Rightarrow a = 2\sqrt{2}$$

(taking the positive root, since a must lie between 2 and 4).

Final Answer

$$a = 2\sqrt{2}$$

Common Mistake: Forgetting to check that the solved value of a actually lies inside the original interval $[2, 4]$ – always sanity-check the root against the region's bounds ($2\sqrt{2} \approx 2.83$, which does lie in $[2, 4]$, confirming it's the valid root).

Quick Recall: "Equal-area division" is a two-step problem: find the total area, then set the partial-area integral (with the unknown as a limit) equal to half of it, and solve.

Type 25: Ratio of two areas from the same curve family

Two regions are defined using the *same* curve, cut off at different ordinates, and possibly with different quadrant restrictions. Compute each area separately (paying close attention to whether each region is restricted to one branch or includes both), then simplify the ratio.

Formula Used: $A = \int_0^{x_0} 2\sqrt{kx} \, dx = \frac{4}{3}\sqrt{k} x_0^{3/2}$

Source: Exercise 8, Q36

Given: $A_1 : y^2 = 4x, x = 1, \text{ x-axis, first quadrant only}$ $A_2 : y^2 = 4x, x = 4 \text{ (no quadrant restriction stated)}$

Solution:

A_1 is a single-branch (first-quadrant) region:

$$A_1 = \int_0^1 2\sqrt{x} \, dx = \left[\frac{4}{3}x^{3/2} \right]_0^1 = \frac{4}{3}$$

Since A_2 is *not* restricted to the first quadrant, both branches of the parabola (above and below the x-axis) are included, so it is double the single-branch computation.

$$A_2 = 2 \int_0^4 2\sqrt{x} \, dx = 2 \left[\frac{4}{3}x^{3/2} \right]_0^4 = 2 \times \frac{4}{3} \times 8 = \frac{64}{3}$$

$$A_1 : A_2 = \frac{4}{3} : \frac{64}{3} = 4 : 64 = 1 : 16$$

Final Answer

$$A_1 : A_2 = 1 : 16$$

Common Mistake: Missing the wording difference – A_1 explicitly says "first quadrant" while A_2 does not, which quietly doubles A_2 ; skimming past this detail gives the wrong ratio (1:8 instead of the correct 1:16).

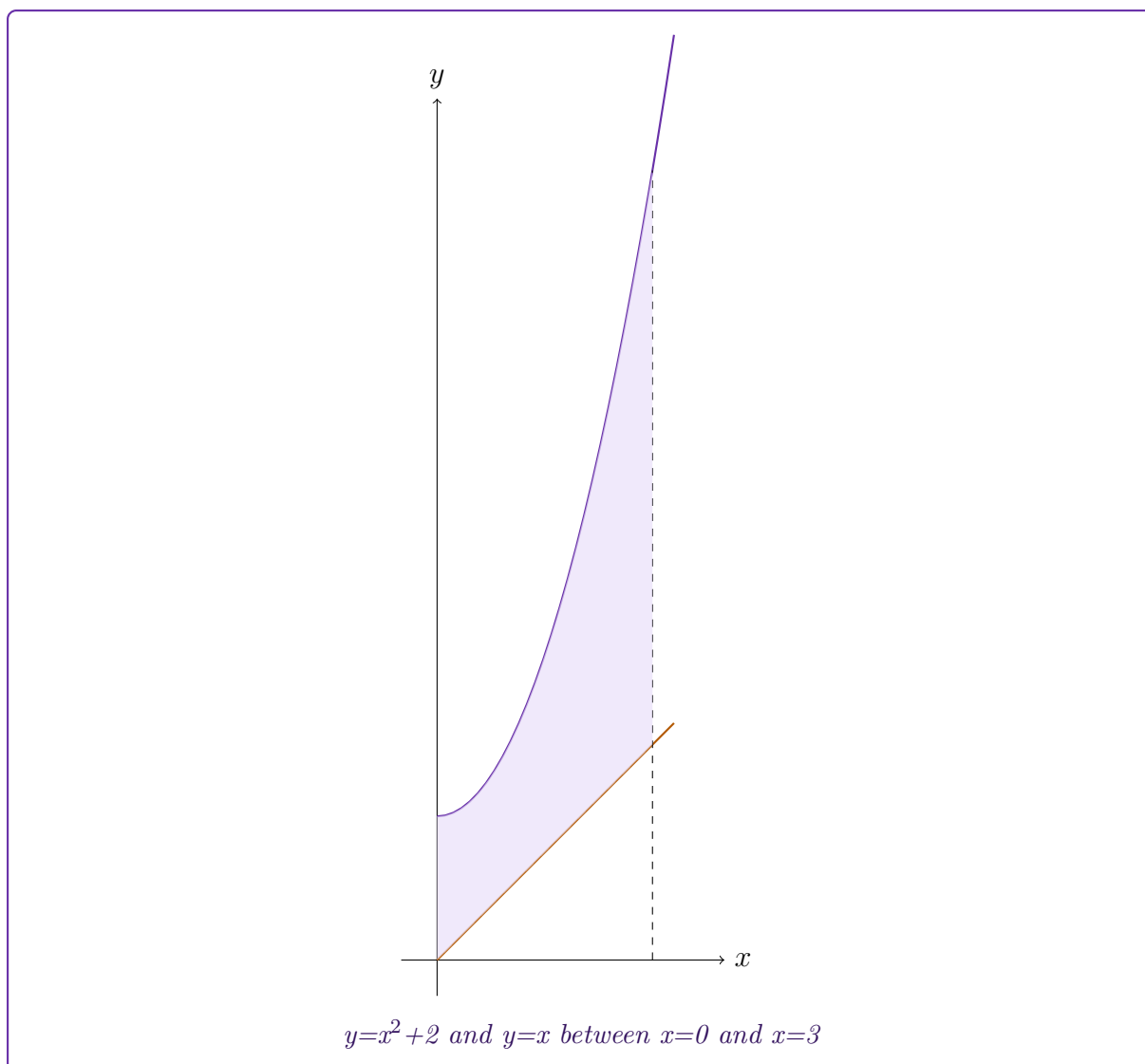
Quick Recall: Before computing a ratio question, re-read each region's description separately – identical-looking curve conditions can still describe different amounts of area if the quadrant restriction differs.

Type 26: Region bounded by $y=x^2+2$ and two lines

A parabola shifted upward and a straight line that never meets it (check the discriminant/complete the square to confirm) – the enclosed region is simply the vertical strip between the two curves, bounded left and right by two given vertical lines, not by their own intersection points.

Formula Used: $A = \int_{x_1}^{x_2} [y_{\text{parabola}} - y_{\text{line}}] dx$

Source: Exercise 8, Q29



Given: $y = x^2 + 2$, $y = x$, $x = 0$, $x = 3$

Solution:

Check whether the curves ever meet: $x^2 + 2 = x \Rightarrow x^2 - x + 2 = 0$, which has discriminant $1 - 8 = -7 < 0$ – no real solutions, so the parabola never touches the line; it stays above it everywhere (confirm with $x = 0$: parabola gives 2, line gives 0). So the region is bounded purely by the two given vertical lines, not by any intersection point.

$$A = \int_0^3 [(x^2 + 2) - x] dx = \left[\frac{x^3}{3} + 2x - \frac{x^2}{2} \right]_0^3$$

$$A = (9 + 6 - 4.5) - 0 = 10.5 = \frac{21}{2}$$

Final Answer

$$A = \frac{21}{2} \text{ sq. units}$$

Common Mistake: Wasting time trying to solve for an intersection point that doesn't exist – always check the discriminant first when two curves look like they "should" cross but the region description instead gives two fixed vertical bounds.

Quick Recall: If two curves never actually cross (check the discriminant), the given vertical lines themselves are the only limits you need – no root-finding required.

Type 27: Region bounded by $y = -x^2$ and a line

A downward parabola and a straight line, both of which dip below the x-axis for part of their range – but since the region asked for is the area *between* the two curves (not between either curve and the x-axis), the sign of y individually doesn't matter; only which curve is on top does.

Formula Used: $A = \int_{x_1}^{x_2} [y_{\text{upper}} - y_{\text{lower}}] dx$

Source: Exercise 8, Q38

Given: $y = -x^2$, $x + y + 2 = 0$ (i.e. $y = -x - 2$)

Solution:

Set the expressions equal: $-x^2 = -x - 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0 \Rightarrow x = -1, 2$. Testing $x = 0$: parabola gives 0, line gives -2 – so the parabola is on top throughout, even though both dip negative elsewhere; only the relative position matters here.

$$\begin{aligned} A &= \int_{-1}^2 [-x^2 - (-x - 2)] dx = \int_{-1}^2 (-x^2 + x + 2) dx \\ &= \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_{-1}^2 \end{aligned}$$

$$\text{At } x = 2: -\frac{8}{3} + 2 + 4 = \frac{10}{3}. \text{ At } x = -1: \frac{1}{3} + \frac{1}{2} - 2 = -\frac{7}{6}.$$

$$A = \frac{10}{3} - \left(-\frac{7}{6} \right) = \frac{20}{6} + \frac{7}{6} = \frac{27}{6} = \frac{9}{2}$$

Final Answer

$$A = \frac{9}{2} \text{ sq. units}$$

Common Mistake: Trying to split this into "positive part" and "negative part" relative to the x-axis – that splitting is only needed when the region is measured against the x-axis itself, not when it's the area strictly between two curves.

Quick Recall: "Between two curves" only cares which curve is on top – don't confuse this with the "curve crosses the x-axis" splitting rule from Types 2 and 3.

Type 28: ★ Composite region – a circle and $y = \sin x$ (multi-part)

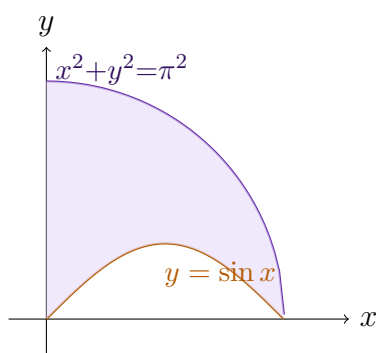
A single question stitches together several skills: finding where a circle and a trigonometric curve meet, verifying a standard integral identity, sketching, and finally computing an enclosed area. Treat each sub-part as its own mini-derivation, in order, since later parts build on earlier ones.

★ Board Favorite – This composite format – intersection, identity check, sketch, area, all on one curve pair – is exactly the style ISC uses for its hardest Long Answer question in this chapter; it is worth practising as a single connected argument rather than four separate questions.

Formula Used: $\int_0^r \sqrt{r^2 - x^2} dx = \frac{\pi r^2}{4}$

Formula Used: $\int \sin x dx = -\cos x + C$

Source: Exercise 8, Q40



Quarter circle $x^2 + y^2 = \pi^2$ and $y = \sin x$, first quadrant

Given: $x^2 + y^2 = \pi^2$, $y = \sin x$, first quadrant

Solution:

Case: (a) Points of intersection

Substitute a candidate: at $x = \pi$, the circle gives $y^2 = \pi^2 - \pi^2 = 0 \Rightarrow y = 0$, and $\sin \pi = 0$ too – so $(\pi, 0)$ satisfies both equations, and is the genuine intersection point of the two curves. The circle also meets the y-axis at $(0, \pi)$; together these two points are the corners that bound the enclosed region used in parts (c) and (d) (only $(\pi, 0)$ is where the two curves themselves actually cross – for every x strictly between 0 and π , the circle stays above the sine curve, as the diagram shows).

Case: (b) Verifying $\int_0^\pi \sqrt{\pi^2 - x^2} dx = \frac{\pi^3}{k}$

Using the standard quarter-circle formula $\int_0^r \sqrt{r^2 - x^2} dx = \frac{\pi r^2}{4}$ with $r = \pi$:

$$\int_0^\pi \sqrt{\pi^2 - x^2} dx = \frac{\pi(\pi^2)}{4} = \frac{\pi^3}{4}$$

Comparing with the given form $\frac{\pi^3}{k}$:

$$k = 4$$

Case: (c) Sketch

See the diagram above – the quarter-circle arc lies entirely above the sine curve for $0 < x < \pi$, touching only at the single point $(\pi, 0)$; the enclosed region is the sliver between them, capped on the left by the y-axis.

Case: (d) Area enclosed by both curves and the y-axis

The enclosed region is the quarter-circle area *minus* the area under the sine curve, over the same interval $[0, \pi]$.

$$A = \int_0^\pi \sqrt{\pi^2 - x^2} dx - \int_0^\pi \sin x dx = \frac{\pi^3}{4} - [-\cos x]_0^\pi$$

$$\int_0^\pi \sin x dx = -\cos \pi - (-\cos 0) = -(-1) + 1 = 2$$

$$A = \frac{\pi^3}{4} - 2$$

Final Answer

(a) $(\pi, 0)$ is the curve intersection; $(0, \pi)$ is the circle's y-axis point (b) $k = 4$
 (d) $A = \frac{\pi^3}{4} - 2$ sq. units

Common Mistake: Assuming the two curves cross somewhere in the interior of the interval just because they "look close" on a rough sketch – always verify algebraically (as in part (a)) rather than trusting the picture alone.

Quick Recall: When a question has parts (a)-(d), each part is a stepping stone – find the intersection first, confirm any given identity next (it usually reveals a constant you'll reuse), sketch to fix the picture in your mind, and only then set up the final area integral.